SpiralMamba: Spatial–Spectral Complementary Mamba With Spatial Spiral Scan for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification is crucial in the remote sensing (RS) community. In recent years, Transformers have been popular in this field due to their global information modeling capabilities. However, the quadratic complexity limits their performance under limited computational resources. Fortunately, a selective structured state space model (SSM) named Mamba emerges. Like Transformer, it is good at modeling the long-distance relationships hidden in the pending data. Unlike Transformer, its complexity remains at a linear level. Therefore, a growing number of studies have been proposed to explore the usefulness of Mamba in HSI classification. Nevertheless, most of them only apply Mamba to HSIs directly but do not consider the inherent characteristics of HSIs properly. To exploit the potential of Mamba in HSI classification deeply, this article presents a new spatial–spectral complementary Mamba with a spatial spiral scan named SpiralMamba. It mainly encloses three main components: a spatial Mamba encoder (SpaME), a spectral Mamba encoder (SpeME), and a spatial–spectral complementary fusion module (SSCFM). SpaME focuses on understanding the spatial context within HSIs. To this end, instead of the common scanning, a spatial spiral scan strategy is introduced to address the sequence transformation of noncausal HSIs. SpeME aims to comprehensively extract valuable spectral features from HSIs. To achieve this goal, besides developing a spectral bidirectional scan strategy, a multilayer convolution (MLC) is also incorporated to capture local variations within spectral tokens. SSCFM concentrates on building the complex connections between spatial and spectral features and fusing them. For this purpose, a relationship learning block (RLB) and a threshold enhancement mechanism (TEM) are developed. Positive experimental results counted on three public HSI datasets demonstrate the effectiveness of SpiralMamba. Our source codes are available at https://github.com/TangXu-Group/Hyperspectral-Images-Classification/tree/main/SpiralMamba**

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***Index Terms*— Hyperspectral image (HSI) classification, selective structured state space model (SSM) (Mamba).**

I. Introduction

HYPERSPECTRAL image (HSI) processing is always a hot topic in the remote sensing (RS) community. Unlike the regular RS products, HSIs contain rich and nearly continuous spectral information [1], [2]. They consist of hundreds of narrow bands with different manifestations on diverse objects [3]. Therefore, HSIs can reflect the physical properties of objects at a fine-grained level and provide richer information for Earth observation [4], [5]. Owing to this property, HSIs can be applied to a wide range of applications, such as agricultural monitoring [6], [7], [8], urban planning [9], [10], environmental monitoring [11], [12], mineral exploration [13], [14], and plants classification [15], [16], [17]. In view of the broad application of HSI, designing a suitable and accurate HSI classification model remains a primary and urgent task.

HSI classification aims to assign the semantic labels to pixels according to their spectral characteristics and spatial structure [18], [19]. In the current deep era, the popular HSI classification methods are mainly developed based on convolutional neural networks (CNNs) [20] and Transformer [21]. The CNN-based methods [22], [23], [24], [25], [26] are good at capturing the local information from HSIs due to their fixed receptive fields. Along with other successful techniques, such as visual attention [27], they achieve good performance. However, since CNN-based approaches cannot fully explore the long-range dependencies hidden in HSIs, their behavior is not as good as expected. In this case, Transformer-based methods [28], [29], [30], [31], [32] stand out for their advantages. With the strong capacities of Transformer in global knowledge modeling, they enhance the HSI classification results to a new degree. Considering the inherent benefits of CNNs and Transformers, researchers intend to combine them for HSI classification [33], [34], [35], [36], [37], [38]. This way, the local and global clues within HSIs can be analyzed thoroughly to ensure the precision of the final classification maps. Although the above methods perform well, they are confronted with an issue, i.e., the quadratic computational complexity of Transformers caused by the self-attention mechanism [39].

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Due to this shortcoming, its generalization in practical applications is limited. Therefore, there is a need to develop an HSI classification method that can discover valuable HSIs clues with a relatively low computational complexity.

Recently, with the emergence and popularity of the selective structured state space model (SSM) (Mamba) [39], [40], a favorable solution to the need mentioned above has become available. Like Transformers, Mamba has strong global modeling capabilities. Unlike Transformers, Mamba's computational complexity stays at the linear level. Especially when vision Mamba [41], [42] comes out, many new Mamba-based models have been proposed and perform well in various computer vision and RS tasks, such as image classification [43], [44], semantic segmentation [45], [46], change detection [47], [48], and so on [49], [50]. Mamba has also attracted scholars in HSI classification [51], [52], [53], [54], [55], [56]. Although their behavior is good, there is still room for improvement. The first challenge is how to convert noncausal HSIs into the sequence for the SSM [39] reasonably. The Mamba-based approaches in [51], [54], [57], [58], [59], and [60] usually leverage simple flatten (sequential scan) or cross scan [42] to achieve HSIs' sequence transformation. Although feasible, they do not consider the spatial characteristics of HSI patches that expand around the center pixel, leading to information discontinuity at the edges. As a result, some positional information would be lost, and spatial clues could not be accurately expressed. Second, there is insufficient local modeling within the spectral tokens. Current methods [54], [55], [57], [58] focus on extracting global clues among spectral tokens using Mamba. However, they neglect local correlations within these tokens. Since adjacent features within spectral tokens are often more correlated, the lack of local perception results in incomplete spectral feature extraction. Third, independent spatial/spectral learning or inadequate feature-level fusion. Some Mamba-based methods [53], [56], [57] explore spatial or spectral messages from HSIs separately, limiting their ability to understand the HSI data fully. Even some methods [55], [58], [59] take spatial and spectral aspects at the same time, their fusion strategies are often insufficient, which lack deep relational modeling and are sensitive to anomalous conditions, e.g., different spectra for the same material and the same spectra for different materials.

To overcome the challenges mentioned above, we propose a spatial–spectral complementary Mamba with spatial spiral scan for HSI classification named SpiralMamba. It mainly consists of a spatial Mamba encoder (SpaME), a spectral Mamba encoder (SpeME), and a spatial–spectral complementary fusion module (SSCFM). First, under the Mamba framework, we tailor a spatial spiral scan strategy to maintain the spatial structure relative to the center pixel as much as possible. Therefore, SpaME can reduce the information loss in the sequence transition and better understand spatial information. Second, SpeME is introduced to strengthen SpiralMamba from the spectral perspective. In SpeME, rather than multilayer perceptron (MLP), we combine Mamba with multilayer convolution (MLC) to complete the global information interaction and the local feature extraction. This way, the spectral clues hidden in HSIs can be fully and appropriately interpreted. Third, SSCFM is developed to ensure that SpiralMamba can adequately fuse the spatial and spectral features. SSCFM involves a relationship learning block (RLB) and a threshold enhancement mechanism (TEM), which aim to learn the complex relationships between different features and refine the learned relationship weights. The spatial and spectral features can be integrated accordingly in this progressive learning and optimization manner. It is important to note that the choice of Mamba rather than Transformer for HSIs is due not only to its lower computational complexity but also to its inherent inductive bias as a sequence model, which is more suitable for investigating continuous spectral sequences. What is more, since Mamba selectively explores the relationships between the current and historical tokens instead of all tokens [42] and does not have complex attention calculations, it has a lower risk of overfitting on HSIs with limited sample and high spectral dimensions.

Our main contributions are summarized as follows.

1) A new HSI classification network named SpiralMamba, which has a dual-branch architecture, is proposed to comprehensively capture the spatial–spectral information hidden in HSIs. The learned features are then integrated through a delicately designed fusion module. This spatial–spectral complementary cooperation enables the accuracy of the results.
2) In the spatial branch, the spatial spiral scan strategy is developed considering the characteristics of HSIs. Unlike traditional flatten or cross scan, the spiral scan preserves spatial continuity and positional information relative to the center pixel during the sequence transformation. It emphasizes the importance of the center pixel, allowing Mamba to capture spatial information better.
3) Adequate local modeling within spectral tokens in the spectral branch. The introduction of MLC in SpeME enhances local perceptual ability within spectral tokens by leveraging convolutions to capture internal correlations. This local modeling within the token complements Mamba's global information extraction among spectral tokens, leading to a more comprehensive representation of spectral features.
4) Accurate and elaborated feature-level fusion. Unlike simple fusion operations, e.g., addition or concatenation, SSCFM uses RLB to fully model the hidden relationships between spatial and spectral features and applies TEM to refine feature weights further, enhancing reliable features while filtering out unreliable ones to avoid inaccurate fusion.

The rest of this article is structured as follows. Section II reviews existing HSI classification methods, including non-Mamba- and Mamba-based approaches. Section III explains SpiralMamba in detail. The experiments and discussion are exhibited in Section IV. Finally, Section V offers a brief conclusion.

II. Related Work

For clarity, the existing HSI classification methods would be roughly categorized into two groups according to whether

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there is a Mamba or not, and we record them as non-Mamba- and Mamba-based methods.

A. Non-Mamba-Based Methods

In the last few years, CNNs have dominated the HSI classification community due to their powerful feature representation ability. Early CNN-based methods leverage 2-D convolutions to mine local spatial information from HSIs. For example, a contextual deep CNN was proposed in [61], which embeds the residual connections into a normal CNN to deepen and widen the network. This way, the discriminative local features can be obtained using the limited training samples. Considering the specific structure of HSIs (i.e., 3-D cube data), Zheng et al. [62] developed a mixed CNN network in which the 2-D and 3-D convolutions are used simultaneously. This model mitigates the shortcomings of 2-D convolution in exploring spectral knowledge and greatly improves the HSI classification results. Later, with the boom of the attention mechanism [63], an increasing number of CNN-based methods have been presented with the help of various attention mechanisms. For instance, Hang et al. [64] proposed an attention-aided CNN network. It respectively adopts the spatial and spectral attention module to help CNN capture the discriminative spatial positions and bands. Thus, more accurate classification results can be obtained.

Although CNN-based methods perform well, the long-distance information hidden in HSIs cannot be fully explored, which limits their behavior more or less. To overcome this problem, Transformer [21], [27] enters into the field of HSI classification. Benefiting from its powerful global modeling capabilities, Transformer has quickly gained popularity in HSI classification tasks, and Transformer-based methods have also become the mainstream approaches. Hong et al. [65] proposed a network called SpectralFormer to complete HSI classification. It may be the first Transformer-based method, and its contributions lie in breaking the limitation of the receptive field of traditional CNNs and using multihead self-attention to fully model the long-distance dependency between spectra. Another network named morphological Transformer (MorphFormer) was developed under the Transformer architecture in [66]. It first combines the dilation and erosion operations to construct a morphological convolution to explore the rich local clues. Then, along with the self-attention mechanism, MorphFormer can better realize the information interaction of structure and shape to enhance the results. To further investigate the spatial–spectral information within HSIs, Jiang et al. [67] introduced graph generative structure-aware Transformer (GraphGST) with the assistance of Transformer. Instead of using HSI patches or pixel embeddings, GraphGST adopts the graph to generate the tokens for the multihead self-attention. In this fashion, the non-Euclidean structural relationships between tokens can be captured, improving the discrimination of spatial–spectral features. Unlike the above methods, the method published in [68] reconsiders the role of central pixels in HSIs and proposes a relative position encoding scheme to avoid information interference caused by absolute position. Based on this scheme, a spectral–spatial center attention (SSCA) is developed, which uses the spectral features in the center pixel to query the spatial features in adjacent pixels for more accurate classification.

Recognizing the unique advantages of CNN and Transformer, researchers tend to combine them for HSI classification. Such methods have local and global modeling abilities, which is good for comprehensively extracting spatial–spectral features. For example, Xu et al. [69] proposed a dual-branch network named double branch convolution-transformer network (DBCTNet), where CNN and Transformer are combined in parallel. On the one hand, a convolution-improved Transformer encoder captures long-distance dependencies. On the other hand, a 3-D-CNN is used to model local spatial–spectral information. Both global and local information are taken into account to achieve more accurate classification results. Similarly, a convolutional transformer dual-branch network with efficient interactive self-attention (EISA) was introduced in [70] named dual-branch convolutional transformer network (DCTN). The Transformer branch uses efficient interactive selfattention (ESIA) to interactively fuse the features of height, width, and depth to fully extract global features, while the CNN branch uses 2-D convolution to enhance local features. In this way, the accuracy of HSI classification is further improved.

B. Mamba-Based Methods

Recently, Mamba [39], [40] has become an effective alternative to Transformer due to its advantages in the global receptive field and linear complexity. Many scholars are attempting to apply Mamba to the HSI classification. The first work is an HSIMamba proposed in [57]. It designs a bidirectional SSM to model the global spectral sequences with lower computational complexity, breaking through the dilemma of Transformer under limited computing resources. Yao et al. [56] developed another Mamba-based method named SpectralMamba. It uses a piecewise sequential scanning (PSS) strategy to divide high-dimensional spectral bands into shorter sequences, enabling SSM to fully explore the spectrum's local information and long-term context. In addition, while reducing the computational scale, it also breaks the dimensionality curse of HSIs. The above methods mainly focus on modeling spectral features. In contrast, Mamba-in-Mamba architecture (MiM) [53] pays more attention to spatial feature extraction. It first develops a centralized Mamba cross-scan scheme based on cross scan to help Mamba better understand the spatial context. Then, a tokenized Mamba (T-Mamba) encoder with a Gaussian decay mask (GDM), a semantic token learner (STL), and a semantic token fuser (STF) are created and integrated to improve the classification accuracy further.

Unlike the mentioned methods that only apply Mamba to explore the spatial or spectral messages from HSIs, some other works use Mamba to extract features from space and spectrum. For example, Wang et al. [54] proposed a spatial-spectral state space model (S^2Mamba) for HSI classification. It first uses Mamba to explore long-range dependencies in space and spectrum dimensions. Then, a spatial–spectral mixture gate is

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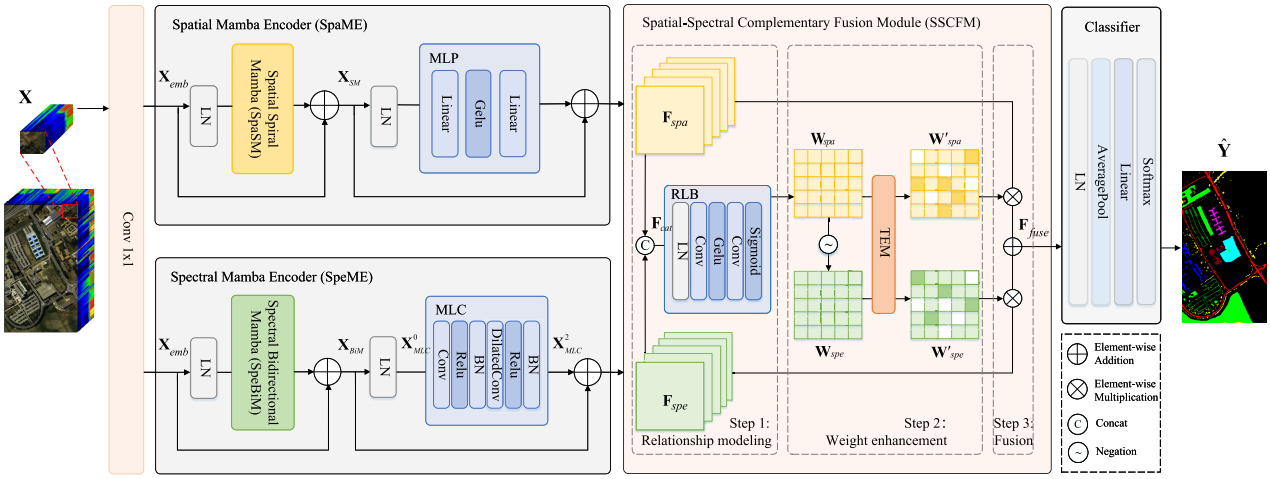


Fig. 1. Overall framework of SpiralMamba. It primarily consists of three components: a SpaME, a SpeME, and an SSCFM. An HSI patch $\mathbf{X}$ is initially embedded using a 1×1 convolution. Then, SpaME and SpeME are used to encode the spatial contextual information and the spectral long-range dependencies, resulting in spatial and spectral features $\mathbf{F}_{\text{spa}}$ and $\mathbf{F}_{\text{spe}}$, respectively. Next, they are simultaneously input into SSCFM to obtain the spatial–spectral fused feature $\mathbf{F}_{\text{fuse}}$. Finally, a simple classifier is used to predict the final classification results.

constructed to fuse the extracted spatial–spectral features adaptively. Using these valuable features, satisfactory classification maps can be produced. For another example, a spectral–spatial Mamba (SS-Mamba) was presented in [55], which also focuses on learning useful spatial–spectral features from HSIs. An SS-Mamba block (SS-MB) is developed to mine and interact with the spatial and spectral information. By stacking multiple SS-MBs, the high-level features can be extracted. Unlike normal Mamba-based networks, SS-Mamba uses position encoding instead of a scanning mechanism to ensure position information during sequence transition. This can avoid considering the scanning order, and the combination with spatial and spectral tokens can also achieve accurate information representation. Not surprisingly, combining CNN and Mamba is also a helpful solution for HSI classification. Sheng et al. [51] constructed a DualMamba, where a lightweight CNN and Mamba are incorporated in a parallel architecture. DualMamba uses the Mamba branch to model the spatial and spectral features from the global aspect and leverages the CNN branch to improve the local perception ability. This way, the HSI classification results are further improved.

III. Method

A. Overall Framework

The overall architecture of SpiralMamba is shown in Fig. 1, mainly enclosing three parts: a SpaME, a SpeME, and an SSCFM. Consistent with the usual HSI classification networks [65], [71], [72], our SpiralMamba's inputs are the HSI patches. For an HSI patch $\mathbf{X} \in \mathbb{R}^{P \times P \times B}$ (where P and B represent the patch size and the number of spectral bands, respectively), SpiralMamba first uses a 1×1 convolution to embed it into $\mathbf{X}_{\text{emb}} \in \mathbb{R}^{P \times P \times N}$, where N means the feature dimension. This step can maintain spatial information and help the model better process and understand high-dimensional HSIs. Then, SpaME and SpeME are respectively applied on $\mathbf{X}_{\text{emb}}$ to fully extract spatial context information and long-range spectral dependencies and generate the spatial feature $\mathbf{F}_{\text{spa}} \in \mathbb{R}^{P \times P \times N}$ and spectral feature $\mathbf{F}_{\text{spe}} \in \mathbb{R}^{P \times P \times N}$. Next, SSCFM is used to fuse $\mathbf{F}_{\text{spa}}$ and $\mathbf{F}_{\text{spe}}$ with the consideration of their hidden relationships. When we get the fused spatial–spectral feature $\mathbf{F}_{\text{fuse}} \in \mathbb{R}^{P \times P \times N}$, a simple classifier is adopted to produce the final classification results.

B. Spatial Mamba Encoder

SpaME aims to comprehensively explore discriminative spatial features. As shown in Fig. 1, SpaME consists of a spatial spiral Mamba (SpaSM), an MLP (composed of two linear layers and a GELU layer [73]), two layer normalization (LN) layers, and two residual modules. The whole procedure of SpaME can be formulated as follows:

$$\begin{aligned} \mathbf{X}_{\text{SM}} &= \text{SpaSM}(\text{LN}(\mathbf{X}_{\text{emb}})) + \mathbf{X}_{\text{emb}} \\ \mathbf{F}_{\text{spa}} &= \text{MLP}(\text{LN}(\mathbf{X}_{\text{SM}})) + \mathbf{X}_{\text{SM}} \end{aligned} \tag{1}$$

where $\mathbf{X}_{\text{SM}}$ and $\mathbf{F}_{\text{spa}}$ indicate the outputs of SpaSM and SpaME, and SpaSM$(\cdot)$, MLP$(\cdot)$, and LN$(\cdot)$ denote the functions of SpaSM, MLP, and LN, respectively. Among them, SpaSM is responsible for mining the contextual relationship between spatial tokens, while MLP focuses on modeling the spatial features within the spatial tokens. In this way, SpaME to fully understand the spatial information.

The core part of SpaME is SpaSM, which is developed based on an improved Mamba [42] since it is more suitable for modeling 2-D visual data. The flowchart of SpaSM is displayed in Fig. 2, whose basic components are two linear layers, a depthwise convolution (DWConv) layer [74], a sigmoid linear unit (SiLU) layer [75], an SSM layer [39], and an LN layer. Besides, we develop a spatial spiral scan strategy and a spatial spiral merge scheme, and embed them before/after the SSM layer. For clarity, we will refer to them as spiral scan and spiral merge in the following explanation. Unlike usual flatten or cross scan, the spiral scan fully considers the characteristics

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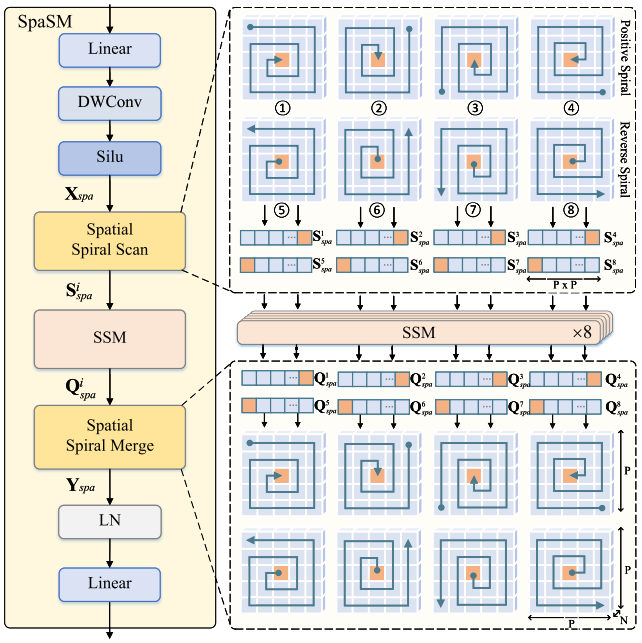


Fig. 2. Flowchart of the SpaSM. The input data is first passed through a linear layer, a DWConv layer, and a SiLU layer to obtain 2-D features $\mathbf{X}_{\text{spa}}$. Then, the spatial spiral scan is used to convert $\mathbf{X}_{\text{spa}}$ into eight different 1-D spatial sequences. The spatial clues can be explored using these sequences along with eight unshared SSMs. Next, the spiral merge is used to restore the processed sequences into 2-D versions and combine them. Finally, the output of SpaSM can be obtained by passing the restored and integrated feature through an LN layer and a linear layer. ①–⑧ in the figure refer to the eight different spatial spiral scan paths.

of HSI patches and unfolds along the central pixel. It can ensure the position of the neighborhood pixels relative to the central pixel as much as possible during the sequence transition, which avoids the SSM's misunderstanding of spatial information due to the loss of position information. In addition, the spatial sequence obtained by spiral scan will not have the problem of abrupt changes in edge pixels, and spatially adjacent pixels are still adjacent in the sequence, which helps SSM better capture spatial contextual relationships.

As shown in Fig. 2, spiral scan uses eight different spiral scanning paths to convert 2-D spatial features into eight 1-D spatial sequences with different orders. Suppose $\mathbf{X}_{\text{emb}}$ has been mapped into $\mathbf{X}_{\text{spa}} \in \mathbb{R}^{P\times P\times N}$ by passing through the LN, linear, DWConv, and SiLU layers. The definition of $\mathbf{X}_{\text{spa}}$ is

$$\mathbf{X}_{\text{spa}} = \begin{pmatrix} \mathbf{x}_{1,1} & \mathbf{x}_{1,2} & \cdots & \mathbf{x}_{1,P} \\ \mathbf{x}_{2,1} & \mathbf{x}_{2,2} & \cdots & \mathbf{x}_{2,P} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}_{P,1} & \mathbf{x}_{P,2} & \cdots & \mathbf{x}_{P,P} \end{pmatrix} \tag{2}$$

where $\mathbf{x}_{i,j} \in \mathbb{R}^N$ represents the spectral features of the pixel in the ith row and jth column. The central pixel of $\mathbf{X}_{\text{spa}}$ can be represented as $x_{c,c} \in \mathbb{R}^N$, where $c = ((P+1)/2)$. The eight spiral paths of spiral scan for $\mathbf{X}_{\text{spa}}$ and the generated eight 1-D spatial sequences $\{\mathbf{S}^i{}_{\text{spa}}, i = 1, \ldots, 8\} \in \mathbb{R}^{P^2\times N}$ are summarized as follows.

1) *Path ①:* Starting from the upper left corner pixel $\mathbf{x}_{1,1}$, scan each pixel in a clockwise spiral until the scan ends at the center pixel $\mathbf{x}_{c,c}$. Thus, the first 1-D sequence $\mathbf{S}^1_{\text{spa}} = [\mathbf{x}_{1,1}, \mathbf{x}_{1,2}, \ldots, \mathbf{x}_{c,c}]$ is obtained.
2) *Paths ②–④:* Same as path ①, except that the starting point is replaced by the upper right corner pixel $\mathbf{x}_{1,P}$, the lower left corner pixel $\mathbf{x}_{P,1}$, and the lower right corner pixel $\mathbf{x}_{P,P}$, respectively. The endpoint is still the center pixel $\mathbf{x}_{c,c}$. Thus, the second to fourth 1-D sequences $\mathbf{S}^2_{\text{spa}} = [\mathbf{x}_{1,P}, \mathbf{x}_{2,P}, \ldots, \mathbf{x}_{c,c}]$, $\mathbf{S}^3_{\text{spa}} = [\mathbf{x}_{P,1}, \mathbf{x}_{P,2}, \ldots, \mathbf{x}_{c,c}]$, and $\mathbf{S}^4_{\text{spa}} = [\mathbf{x}_{P,P}, \mathbf{x}_{P,P-1}, \ldots, \mathbf{x}_{c,c}]$ are obtained.
3) *Paths ⑤–⑧:* These four paths are the reversal of the previous paths ①–④, forming a counterclockwise spiral path with the center pixel as the starting point and the four vertex pixels $\mathbf{x}_{1,1}$, $\mathbf{x}_{1,P}$, $\mathbf{x}_{P,1}$, and $\mathbf{x}_{P,P}$ as the endpoints. Thus, $\mathbf{S}^5_{\text{spa}} = [\mathbf{x}_{c,c}, \ldots, \mathbf{x}_{1,2}, \mathbf{x}_{1,1}]$, $\mathbf{S}^6_{\text{spa}} = [\mathbf{x}_{c,c}, \ldots, \mathbf{x}_{2,P}, \mathbf{x}_{1,P}]$, $\mathbf{S}^7_{\text{spa}} = [\mathbf{x}_{c,c}, \ldots, \mathbf{x}_{P,2}, \mathbf{x}_{P,1}]$, and $\mathbf{S}^8_{\text{spa}} = [\mathbf{x}_{c,c}, \ldots, \mathbf{x}_{P,P-1}, \mathbf{x}_{P,P}]$ are obtained.

The reasons for setting the scanning path this way are summarized as follows.

1) *Emphasizing the Importance of the Center Pixel:* Due to Mamba's sequential processing mechanism, it tends to focus more on the end of the sequence. The positive spiral paths (i.e., ①–④) end at the center pixel, ensuring that Mamba pays more attention to the most crucial position (the center pixel) and captures the HSI patch's core information.
2) *Bidirectional Context Capture:* The positive spiral paths traverse in the clockwise direction, while the reverse spiral paths (i.e., ⑤–⑧) follow the counterclockwise direction. By combining them, SpaSM can capture the spatial context in both directions, free from the constraints of a single direction.
3) *Preserving Spatial Continuity:* Flatten operation and cross scan often disturb the spatial clues at edge pixels. The proposed spiral scan ensures the continuity of spatial information, avoiding sudden discontinuities.
4) *Maintain the Positions Relative to the Center Pixel:* HSI patches are generated according to the center pixel. Our spiral scan can preserve neighboring pixels' positions relative to the center pixel throughout the sequence conversion. This minimizes the loss of positional information that could lead to misinterpretation by the model.

When we get various sequences, we will input them into eight unshared SSMs to model the relationship between spatial pixels. This step can be expressed as follows:

$$\begin{aligned} \mathbf{h}^i_j &= \bar{\mathbf{A}}_{\text{spa}}\mathbf{h}^i_{j-1} + \bar{\mathbf{B}}_{\text{spa}}\left(\mathbf{S}^i_{\text{spa}}\right)_j \\ \left(\mathbf{Q}^i_{\text{spa}}\right)_j &= \mathbf{C}_{\text{spa}}\mathbf{h}^i_j + \left(\mathbf{S}^i_{\text{spa}}\right)_j \end{aligned} \tag{3}$$

where $\bar{\mathbf{A}}_{\text{spa}}$, $\bar{\mathbf{B}}_{\text{spa}}$, and $\mathbf{C}_{\text{spa}}$ represent the learnable parameters in SSM, $\mathbf{h}^i_j$ represents the jth hidden state of the ith sequence, and $\mathbf{Q}^i_{\text{spa}} \in \mathbb{R}^{P^2\times N}$ implies the output of SSMs. In this fashion, the spatial clues hidden in HSIs can be fully analyzed from the global aspects.

To facilitate subsequent feature processing, $\{\mathbf{Q}^i{}_{\text{spa}}, i = 1, \ldots, 8\}$ should be restored into 2-D version and inte-

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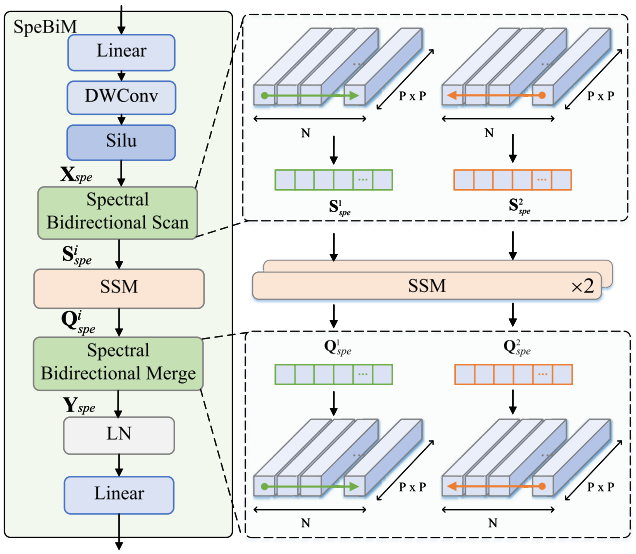


Fig. 3. Flowchart of the SpeBiM. The input features are first processed by the linear, DWConv, and SiLU layers to generate $\mathbf{X}_{\text{spe}}$. Then, a spectral bidirectional scan is applied on $\mathbf{X}_{\text{spe}}$ to convert them into two 1-D spectral sequences in forward and reverse directions. The long-range spectral dependencies can be mined by two unshared SSMs. Next, a spectral bidirectional merge combines sequences and reshapes them into a 2-D version. Finally, the output of SpeBiM can be obtained through an LN layer and a linear layer.

grated. To this end, we develop a spiral merge. Specifically, the eight sequences are first returned along their respective scanning paths to achieve eight 2-D features with the original spatial shape. Then, we integrate them into a spatial feature $\mathbf{Y}_{\text{spa}} \in \mathbb{R}^{P\times P\times N}$ by adding them element by element.

C. Spectral Mamba Encoder

SpeME aims to extract complex spectral features from the global and local perspectives. As shown in Fig. 1, SpeME consists of a spectral bidirectional Mamba (SpeBiM), an MLC, two LN layers, and two residual modules. Its formula is expressed as follows:

$$\begin{aligned}\mathbf{X}_{\text{BiM}} &= \text{SpeBiM}(\text{LN}(\mathbf{X}_{\text{emb}})) + \mathbf{X}_{\text{emb}}\\ \mathbf{F}_{\text{spe}} &= \text{MLC}(\text{LN}(\mathbf{X}_{\text{BiM}})) + \mathbf{X}_{\text{BiM}}\end{aligned} \tag{4}$$

where $\mathbf{X}_{\text{BiM}}$ and $\mathbf{F}_{\text{spe}}$ represent the outputs of SpeBiM and SpeME, and SpeBiM$(\cdot)$, MLC$(\cdot)$, and LN$(\cdot)$ denote the functions of SpeBiM, MLC, and LN, respectively. Here, SpeBiM is used to model the global long-term dependencies between spectral tokens, and MLC focuses on extracting local features within spectral tokens. With the cooperation of SpeBiM and MLC, SpeME can encode high-dimensional spectral features comprehensively and accurately.

The structure of SpeBiM is shown in Fig. 3. Like SpaME, it mainly encloses two linear layers, a DWConv layer, a SiLU layer, an SSM layer, and an LN layer. Before and after the SSM layer, a spectral bidirectional scan and a spectral bidirectional merge are embedded. For simplicity, we refer to them as bidirectional scan and bidirectional merge. Suppose $\mathbf{X}_{\text{emb}}$ has been mapped into $\mathbf{X}_{\text{spe}} \in \mathbb{R}^{P\times P\times N}$ by the LN, linear, DWConv, and SiLU layers. Since the spectral bands of HSIs are continuous, the spectral features can be regarded as sequences. Therefore, the bidirectional scan first flattens $\mathbf{X}_{\text{spe}}$ along the spatial dimension. After the forward and reverse scanning, two different spectral sequences $\{\mathbf{S}^1_{\text{spe}}, \mathbf{S}^2_{\text{spe}}\} \in \mathbb{R}^{N\times P^2}$ can be achieved. Then, the two spectral sequences obtained are input into two unshared SSMs to extract the contextual information between spectra from a global perspective. The following formula can express this process:

$$\begin{aligned}\tilde{\mathbf{h}}^i_j &= \bar{\mathbf{A}}_{\text{spe}}\tilde{\mathbf{h}}^i_{j-1} + \bar{\mathbf{B}}_{\text{spe}}\left(\mathbf{S}^i_{\text{spe}}\right)_j\\ \left(\mathbf{Q}^i_{\text{spe}}\right)_j &= \mathbf{C}_{\text{spe}}\tilde{\mathbf{h}}^i_j + \left(\mathbf{S}^i_{\text{spe}}\right)_j\end{aligned} \tag{5}$$

where $\tilde{\mathbf{h}}^i_j$ represents the jth hidden state of the ith sequence, $\bar{\mathbf{A}}_{\text{spe}}$, $\bar{\mathbf{B}}_{\text{spe}}$, and $\mathbf{C}_{\text{spe}}$ represent the learnable parameters of SSM, and $\{\mathbf{Q}^1_{\text{spe}}, \mathbf{Q}^2_{\text{spe}}\}$ represent the output sequences of the SSMs. Next, bidirectional merge is used to reverse $\mathbf{Q}^2_{\text{spe}}$ and add it to $\mathbf{Q}^1_{\text{spe}}$ at element level, and then reshape the summation result to $\mathbf{Y}_{\text{spe}} \in \mathbb{R}^{P\times P\times N}$. Finally, $\mathbf{Y}_{\text{spe}}$ passes through an LN layer, a linear layer, and a residual connection to obtain the SpeBiM output $\mathbf{X}_{\text{BiM}}$.

When we get $\mathbf{X}_{\text{BiM}}$, MLC instead of MLP is used to refine it by analyzing the value messages within spectral tokens. MLC involves a 3×3 convolution layer and a 3×3 dilated convolution layer. Also, each comes with a ReLU layer and a batch normalization (BN) layer. Assume that $\mathbf{X}_{\text{BiM}}$ has been mapped into $\mathbf{X}^0_{\text{MLC}}$ through an LN layer. $\mathbf{X}^0_{\text{MLC}}$ is first passed through the 3×3 convolution layer with ReLU and BN to obtain the intermediate feature $\mathbf{X}^1_{\text{MLC}}$. Next, $\mathbf{X}^1_{\text{MLC}}$ is passed through the 3×3 dilated convolution with ReLU and BN further to capture discriminative information in a larger receptive field and achieve the output of MLC $\mathbf{X}^2_{\text{MLC}}$.

D. Spatial–Spectral Complementary Fusion Module

The target of SSCFM is to fully integrate spatial and spectral features extracted by two Mamba Encoders. As shown in Fig. 1, this module can be divided into three steps, i.e., relationship modeling, weight enhancement, and fusion.

There is an RLB in the relationship modeling step, which aims to mine the complex spatial–spectral relationships between $\mathbf{F}_{\text{spa}} \in \mathbb{R}^{P\times P\times N}$ and $\mathbf{F}_{\text{spe}} \in \mathbb{R}^{P\times P\times N}$. After concatenating $\mathbf{F}_{\text{spa}}$ and $\mathbf{F}_{\text{spe}}$ into $\mathbf{F}_{\text{cat}} \in \mathbb{R}^{P\times P\times 2N}$, the relationship weight $\mathbf{W} \in \mathbb{R}^{P\times P}$ can be obtained by the RLB, which consists of two 3×3 convolutions, an LN layer, a GELU layer, and a sigmoid layer. Here, $\mathbf{W}$ can be regarded as the spatial weight $\mathbf{W}_{\text{spa}} \in \mathbb{R}^{P\times P}$. Thus, $1-\mathbf{W}$ can be deemed as the spectral weight $\mathbf{W}_{\text{spe}} \in \mathbb{R}^{P\times P}$. The following formula can represent the above process:

$$\begin{aligned}\mathbf{W} &= \text{RLB}\left(\text{Cat}\left(\mathbf{F}_{\text{spa}}, \mathbf{F}_{\text{spe}}\right)\right)\\ \mathbf{W}_{\text{spa}} &= \mathbf{W}, \mathbf{W}_{\text{spe}} = 1-\mathbf{W}\end{aligned} \tag{6}$$

where RLB$(\cdot)$ represents the function of RLB, and Cat$(\cdot)$ implies the concatenation operation.

In the weight enhancement step, considering the potential anomalies of same-object with different-spectral or same-spectral with different-object situations in HSIs, not all features are trustworthy. Therefore, TEM is applied to refine the spatial and spectral weights, which can be expressed

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as follows:

$$\left(\mathbf{W}'_{\text{spa}}\right)_{ij} = \begin{cases} 1, & \left(\mathbf{W}_{\text{spa}}\right)_{ij} > \lambda \\ 0, & \left(\mathbf{W}_{\text{spa}}\right)_{ij} < 1-\lambda \\ \left(\mathbf{W}_{\text{spa}}\right)_{ij}, & \text{otherwise} \end{cases}$$

$$\left(\mathbf{W}'_{\text{spe}}\right)_{ij} = \begin{cases} 1, & \left(\mathbf{W}_{\text{spe}}\right)_{ij} > \lambda \\ 0, & \left(\mathbf{W}_{\text{spe}}\right)_{ij} < 1-\lambda \\ \left(\mathbf{W}_{\text{spe}}\right)_{ij}, & \text{otherwise} \end{cases} \tag{7}$$

where $(\mathbf{W}_{\text{spa}})_{ij}$, $(\mathbf{W}_{\text{spe}})_{ij}$, $(\mathbf{W}'_{\text{spa}})_{ij}$, and $(\mathbf{W}'_{\text{spe}})_{ij}$ indicate the values of the ith row and jth column in the spatial and spectral weights before and after refinement, and λ denotes a threshold with a value range of [0.5, 1]. In this way, lower values in the relationship weight can be filtered, and higher values can be enhanced, which allows our model to trust the party with higher credibility and eliminates redundant erroneous information.

In the fusion step, the refined relationship weights $\mathbf{W}'_{\text{spa}} \in \mathbb{R}^{P\times P}$ and $\mathbf{W}'_{\text{spe}} \in \mathbb{R}^{P\times P}$ are used to perform a weighted sum of the spatial and spectral features for the fused feature $\mathbf{F}_{\text{fuse}} \in \mathbb{R}^{P\times P\times N}$, which can be defined as follows:

$$\mathbf{F}_{\text{fuse}} = \mathbf{W}'_{\text{spa}} \cdot \mathbf{F}_{\text{spa}} + \mathbf{W}'_{\text{spe}} \cdot \mathbf{F}_{\text{spe}}. \tag{8}$$

E. Classifier

At the top of SpiralMamba, a simple classifier is used to generate the final classification probability distribution $\hat{\mathbf{Y}}$ by using the fused features $\mathbf{F}_{\text{fuse}}$. As shown in Fig. 1, the classifier consists of an LN layer, an average pooling layer, a linear layer, and a softmax layer. This article adopts the ordinary cross-entropy loss to supervise model learning. Its formula is

$$L_{\text{ce}}\left(\mathbf{Y}, \hat{\mathbf{Y}}\right) = -\sum_{i=1}^{C} \mathbf{Y}_i \log\left(\hat{\mathbf{Y}}_i\right) \tag{9}$$

where C is the number of class, $\mathbf{Y}$ is the ground truth, and $\hat{\mathbf{Y}}_i$ is the probability of category i.

IV. Experiment

A. Datasets

The following four datasets are selected to verify our SpiralMamba, including Indian Pines (IP),[1] Pavia University (PU),[1] Houston 2013 (HUS2013),[2] and Houston 2018 (HUS2018).[3]

1) *IP:* It is a vegetation coverage map of Indiana, USA, which contains 16 different vegetation categories. The collected HSI consists of 145 × 145 pixels and 220 spectral bands. To improve IP's quality and usability, 20 noisy and water absorption bands are removed from the original HSI. Fig. 4 shows its false-color image and the ground truth of the training and testing sets. The division of this dataset is shown in Table I.
2) *PU:* This HSI depicts the land cover of the University of Pavia and surrounding regions, with nine different land object categories. It has 610 × 340 pixels and 115 bands. Twelve noisy bands are removed from the data to enhance PU's usability and quality. The false-color image and the ground truth of the training and testing sets are exhibited in Fig. 5. The data division of PU is shown in Table II.
3) *HUS2013:* This dataset shows the land cover of the University of Houston and the surrounding areas. It contains 15 different landform categories. The collected data contains 349 × 1905 pixels and 144 spectral bands. Fig. 6 presents the false-color image and the ground truth of the training and testing sets. Table III summarizes the specific data division.
4) *HUS2018:* This dataset, sourced from the 2018 IEEE GRSS Data Fusion Contest [76], captures ground objects of Houston city with a 1-m spatial resolution and 48 spectral bands. Its spatial size is 601 × 2384, and 504 856 pixels are labeled with 20 semantic categories. We randomly select 100 samples for each category to construct the training set, and the rest samples are used to establish the testing set. The detailed split is presented in Table IV. Fig. 7 illustrates HUS2018's false-color image and the ground truth for the training and test sets. Compared to the above three datasets, HUS2018 offers a larger scale of ground object information and richer categories.

[1]http://lesun.weebly.com/hyperspectral-data-set.html
[2]http://www.grss-ieee.org/community/technical-committees/data-fusion
[3]https://ieee-dataport.org/open-access/2018-ieee-grss-data-fusion-challenge-%E2%80%93-fusion-multispectral-lidar-and-hyperspectral-data

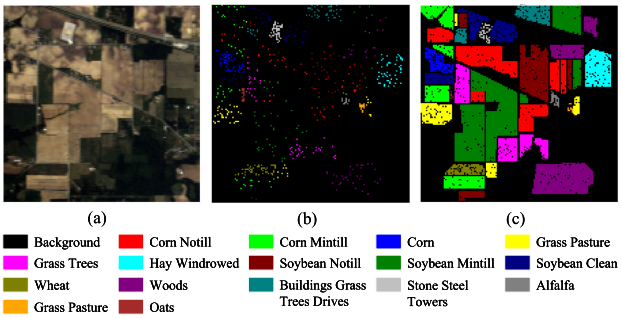


Fig. 4. Examples of the IP dataset. (a) False-color image. (b) and (c) Ground truth of the training and testing sets. The bottom icons show the specific categories represented by each color.

TABLE I
Data Division on IP

Class No.	Class Name	Training	Testing	Total
1	Corn Notill	50	1384	1434
2	Corn Mintill	50	784	834
3	Corn	50	184	234
4	Grass Pasture	50	447	497
5	Grass Trees	50	697	747
6	Hay Windrowed	50	439	489
7	Soybean Notill	50	918	968
8	Soybean Mintill	50	2418	2468
9	Soybean Clean	50	564	614
10	Wheat	50	162	212
11	Woods	50	1244	1294
12	Buildings Grass Trees Drives	50	330	380
13	Stone Steel Towers	50	45	95
14	Alfalfa	15	39	54
15	Grass Pasture Mowed	15	11	26
16	Oats	15	5	20
	Total	695	9671	10366

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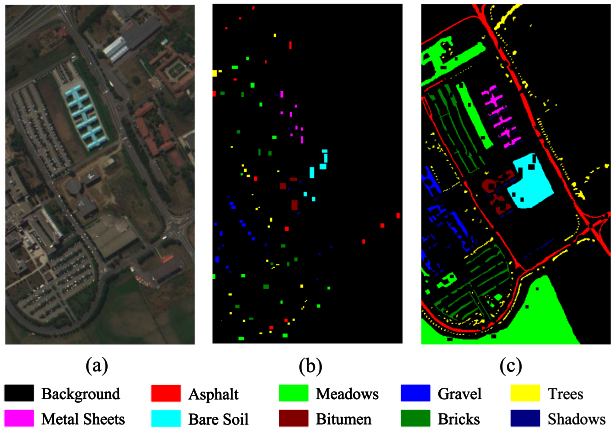


Fig. 5. Examples of the PU dataset. (a) False-color image. (b) and (c) Ground truth of the training and testing sets. The bottom icons show the specific categories represented by each color.

TABLE II
DATA DIVISION ON PU

Class No.	Class Name	Training	Testing	Total
1	Asphalt	548	6304	6852
2	Meadows	540	18146	18686
3	Gravel	392	1815	2207
4	Trees	524	2912	3436
5	Metal Sheets	265	1113	1378
6	Bare Soil	532	4572	5104
7	Bitumen	375	981	1356
8	Bricks	514	3364	3878
9	Shadows	231	795	1026
	Total	3921	40002	43923

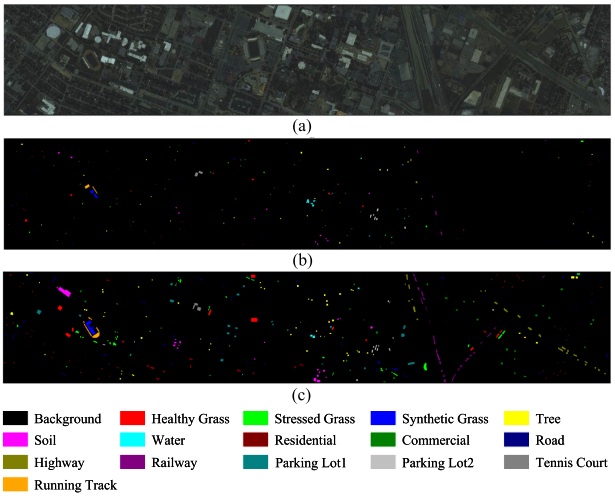


Fig. 6. Examples of the HUS2013 dataset. (a) False-color image. (b) and (c) Ground truth of the training and testing sets. The bottom icons show the specific categories represented by each color.

Note that the data partition schemes for IP, PU, and HUS2013 are in accordance with SprectralFormer [65], and HUS2018 conforms to [77].

B. Implementation Details

1) Experimental Environment and Settings: The experiments are conducted on a GeForce RTX 3090 with 24 GB video memory. The experimental environment is

TABLE III
DATA DIVISION ON HUS2013

Class No.	Class Name	Training	Testing	Total
1	Healthy Grass	198	1053	1251
2	Stressed Grass	190	1064	1254
3	Synthetic Grass	192	505	697
4	Tree	188	1056	1244
5	Soil	186	1056	1242
6	Water	182	143	325
7	Residential	196	1072	1268
8	Commercial	191	1053	1244
9	Road	193	1059	1252
10	Highway	191	1036	1227
11	Railway	181	1054	1235
12	Parking Lot1	192	1041	1233
13	Parking Lot2	184	285	469
14	Tennis Court	181	247	428
15	Running Track	187	473	660
	Total	2832	12197	15029

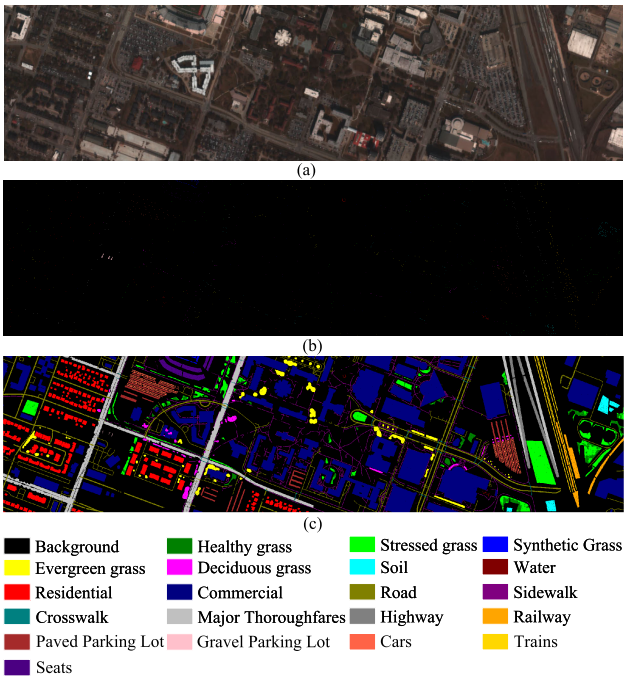


Fig. 7. Examples of the HUS2018 dataset. (a) False-color image. (b) and (c) Ground truth of the training and testing sets. The bottom icons show the specific categories represented by each color.

Python 3.10 and PyTorch 2.1. We adopt the adaptive moment estimation with weight decay (AdamW) algorithm [78] as the optimizer to train our model. The initial learning rate, training epochs, and batch size are set to 0.0001, 500, and 64, respectively. The above settings are consistent on the four datasets. As mentioned in Section III-A, the input data of SpiralMamba is HSI patches. In the following experiments, the patch sizes of IP, PU, HUS2013, and HUS2018 are set to 15, 15, 11, and 17, respectively. In addition, the threshold λ (see Section III-D) should be set in advance. Its empirically optimal values for IP, PU, HUS2013, and HUS2018 are 0.95, 0.90, 0.70, and 0.85, respectively. The influence of patch size and λ will be analyzed in detail in Section IV-E. All experiments are repeated five times randomly to avoid randomness affecting the experimental results. The average results and their variances are reported in this article.

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TABLE IV
DATA DIVISION ON HUS2018

Class No.	Class Name	Training	Testing	Total
1	Healthy grass	100	9699	9799
2	Stressed grass	100	32402	32502
3	Synthetic grass	100	584	684
4	Evergreen grass	100	13495	13595
5	Deciduous grass	100	4921	5021
6	Soil	100	4416	4516
7	Water	100	166	266
8	Residential	100	39672	39772
9	Commercial	100	223652	223752
10	Road	100	45766	45866
11	Sidewalk	100	33929	34029
12	Crosswalk	100	1418	1518
13	Major Thoroughfares	100	46248	46348
14	Highway	100	9765	9865
15	Railway	100	6837	6937
16	Paved Parking Lot	100	11400	11500
17	Gravel Parking Lot	100	46	146
18	Cars	100	6447	6547
19	Trains	100	5269	5369
20	Seats	100	6724	6824
	Total	2000	502856	504856

2) Evaluation Metrics: Three evaluation metrics [69], overall accuracy (OA), average accuracy (AA), and Kappa coefficient, are used to measure the performance of SpiralMamba quantitatively. OA is the overall accuracy of all categories, which aims to evaluate the model's overall performance. AA is the average accuracy of each category, which reflects the average performance of each category and the degree of category imbalance. Kappa coefficient is a measure of observed and expected consistency.

C. Comparison to Existing Methods

Eleven models are selected to comprehensively assess the performance of SpiralMamba, including SpectralFormer [65], MorphFormer [66], spectral query spatial transformer (SQSFormer) [68], GraphGST [67], DBCTNet [69], HSIMamba [57], S^{2}Mamba [54], MiM [53], MambaHSI [58], local-to-global Mamba (MambaLG) [59], and interval group spatial -spectral Mamba framework (IGroupSS-Mamba) [60]. The first five methods are Transformer-based models, in which SpectralFormer, MorphFormer, and SQSFormer are purely developed based on the Transformer, GraphGST is proposed with the help of graph and Transformer, and DBCTNet is constructed with the assistance of the combination of CNN and Transformer. The last six methods are Mamba-based models. All compared approaches are implemented using their open-source codes, and their specific hyperparameters follow the original papers. To be fair, the compared models' data division and experimental environment are consistent with ours, which are discussed in Section IV-B1.

1) Comparison of Numerical Results: Table V presents the results of different methods counted on the IP dataset. We can find that the weakest model is SpectralFormer. This is because it only leverages the vanilla Transformer to complete HSI classification, which can be regarded as an initial attempt to apply Transformers in the HSI community. The Mamba-based models (i.e., S^{2}Mamba, MiM, MambaHSI, and IGroupSS-Mamba) outperform the Transformer-based networks. This happens because SSM in Mamba uses continuous-time dynamic equations to describe the state evolution of the system, which allows Mamba to study the spectral sequence more directly and efficiently and enhances the discriminability of spectral features. However, the performance of HSIMamba and MambaLG does not meet our expectations. The reasons behind this are as follows. HSIMamba explores hidden features in the spectral dimension while neglecting spatial modeling. Although MambaLG can extract spatial and spectral features from local to global contexts, its simple feature fusion scheme leads to the loss of essential features. Although the compared models' results are good, our SpiralMamba still performs best in OA, AA, and Kappa coefficient, with values of 97.76%, 98.76%, and 97.43. Taking OA as an example, our method outperforms SpectralFormer and HSIMamba, which initially explore Transformer and Mamba, by 14.56% and 10.59%. This is because our dual-branch fusion architecture is more effective at capturing spatial–spectral clues. Compared to other Transformer-based models (i.e., MorphFormer, SQSFormer, GraphGST, and DBCTNet), our model surpasses them by 1.25%, 1.5%, 2.87%, and 0.91%. This demonstrates that our designed SpaME and SpeME can accurately encode spatial context and spectral sequences with inherent continuity. Furthermore, compared with S^{2}Mamba, MiM, MambaHSI, MambaLG, and IGroupSS-Mamba, the improvements achieved by SpiralMamba are 0.58%, 0.64%, 1.06%, 6.67%, and 0.78%. This illustrates the usefulness of the proposed spiral scan, facilitating SpiralMamba's ability to capture hidden spatial information and achieve optimal classification results.

Table VI records the experimental results for all models on the PU dataset. As expected, our model achieves the highest OA, AA, and Kappa coefficient scores, which are 98.27%, 97.84%, and 97.66. Besides the positive results mentioned above, there is an encouraging observation: SpiralMamba is the most stable model for this dataset. In contrast, other models exhibit greater variation in performance across different categories. For example, MambaLG performs best in classes 4 and 8 but poorly in classes 1, 2, and 3. For another example, IGroupSS-Mamba can accurately classify the samples in class 5. However, its performance for classes 1 and 3 is weak. By comparison, although our SpiralMamba does not achieve the best in some categories, it consistently maintains a competitive level. These positive results are primarily attributed to the following factors. First, SpiralMamba's comprehensive spatial–spectral modeling plays a key role. Spatially, the eight distinct spiral scan paths cover spatial information from various directions, allowing SpaME to encode spatial features more thoroughly. Spectrally, the complementary cooperation between bidirectional Mamba and MLC captures global bidirectional spectral information between tokens while locally perceiving detailed variations within each token, thus enabling comprehensive spectral feature modeling. Second, the effective spatial–spectral fusion provided by SSCFM further reduces the misclassification. By deeply modeling the relationships within the spatial–spectral information and refining the spatial–spectral weights, SSCFM ensures

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TABLE V

PERFORMANCE OF DIFFERENT METHODS COUNTED ON IP, INCLUDING THE RESULTS OF EACH CLASS AND THE OA, AA, AND KAPPA COEFFICIENT VALUES. THE BEST RESULTS ARE BOLDED (MEAN ± VARIANCE)

Class No.	Transformer-based Methods					Mamba-based Methods						
	SpectralFormer	MorphFormer	SQSFormer	GraphGST	DBCTNet	HSIMamba	S^2Mamba	MiM	MambaHSI	MambaLG	IGroupSS-Mamba	Ours
1	73.48±6.56	94.48±1.36	94.21±1.25	92.54±1.76	94.68±0.46	82.40±2.22	91.85±1.08	**94.90±1.84**	91.60±1.46	32.89±23.5	92.43±2.41	93.84±0.69
2	88.80±1.12	98.44±1.27	97.53±0.42	98.75±0.61	99.03±0.50	76.61±6.49	**99.80±0.10**	98.88±0.22	96.79±0.75	86.58±2.94	98.18±1.24	99.62±0.46
3	91.30±1.24	99.78±0.43	99.89±0.22	98.80±0.94	100.0±0.00	91.09±7.87	100.0±0.00	100.0±0.00	100.0±0.00	86.58±4.18	100.0±0.00	**100.0±0.00**
4	92.66±1.27	97.23±0.90	95.44±0.94	96.51±1.53	95.21±1.06	92.04±1.42	97.45±0.44	96.58±0.93	93.33±0.91	52.93±9.99	98.21±1.44	**99.02±0.39**
5	88.92±1.28	99.60±0.23	99.66±0.23	97.62±2.03	99.17±0.58	96.84±1.77	**100.0±0.00**	99.86±0.16	99.20±0.61	83.80±5.45	98.82±1.56	99.97±0.06
6	95.44±2.07	99.91±0.18	**100.0±0.00**	99.59±0.09	99.82±0.09	98.36±0.53	99.91±0.11	99.91±0.11	99.73±0.09	97.51±1.81	100.0±0.00	99.95±0.09
7	81.09±3.97	91.61±0.66	94.64±2.05	95.99±1.61	94.31±2.42	84.97±3.60	**97.47±0.37**	96.72±0.55	93.92±1.44	0.00±0.00	96.51±2.66	96.32±0.86
8	77.68±2.56	96.25±0.97	93.84±0.91	93.90±3.50	96.44±0.78	80.89±4.22	97.38±0.54	95.58±0.70	95.20±1.20	87.27±1.82	94.51±2.16	**97.88±0.30**
9	72.70±5.14	91.88±1.14	**95.18±0.44**	94.57±1.35	92.70±1.18	88.48±2.47	93.40±1.29	95.13±1.09	91.04±0.52	10.00±20.0	91.36±2.18	93.97±2.34
10	99.63±0.49	100.0±0.00	99.75±0.30	98.64±0.25	100.0±0.00	99.51±0.25	100.0±0.00	100.0±0.00	100.0±0.00	94.27±1.89	99.69±0.31	**100.0±0.00**
11	92.65±2.22	98.65±0.35	98.97±0.20	90.16±18.8	98.73±0.34	96.37±0.76	98.17±0.39	98.79±0.17	99.29±0.29	93.80±2.23	98.77±0.30	**99.66±0.17**
12	83.76±5.69	99.33±0.80	99.88±0.24	98.24±0.82	100.0±0.00	94.18±4.48	99.94±0.12	99.84±0.32	99.76±0.48	94.89±2.25	100.0±0.00	**100.0±0.00**
13	100.0±0.00	100.0±0.00	100.0±0.00	100.00±0.0	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	99.00±0.50	100.0±0.00	**100.0±0.00**
14	71.79±8.11	98.97±1.26	98.97±1.26	100.00±0.0	99.49±1.03	94.36±3.77	100.0±0.00	100.0±0.00	98.46±2.05	98.35±0.66	98.72±1.28	**100.0±0.00**
15	96.36±7.27	100.0±0.00	100.0±0.00	100.00±0.0	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	77.17±3.56	100.0±0.00	**100.0±0.00**
16	100.0±0.00	100.0±0.00	100.0±0.00	100.00±0.0	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	**100.0±0.00**
OA(%)	83.20±1.36	96.51±0.47	96.26±0.35	94.89±3.38	96.85±0.24	87.17±0.37	97.18±0.25	97.12±0.49	96.70±0.15	91.09±0.68	96.98±0.30	**97.76±0.27**
AA(%)	87.89±0.58	97.88±0.31	98.00±0.25	97.21±1.46	98.10±0.16	92.25±0.80	98.46±0.13	98.51±0.28	97.83±0.21	74.69±0.84	98.25±0.43	**98.76±0.19**
$k\times100$	80.84±1.56	96.00±0.54	95.71±0.40	94.16±3.82	96.38±0.28	85.35±0.41	96.76±0.29	96.69±0.56	96.03±0.28	89.87±0.77	96.40±0.35	**97.43±0.31**

TABLE VI

PERFORMANCE OF DIFFERENT METHODS COUNTED ON PU, INCLUDING THE RESULTS OF EACH CLASS AND THE OA, AA, AND KAPPA COEFFICIENT VALUES. THE BEST RESULTS ARE BOLDED (MEAN ± VARIANCE)

Class No.	Transformer-based Methods					Mamba-based Methods						
	SpectralFormer	MorphFormer	SQSFormer	GraphGST	DBCTNet	HSIMamba	S^2Mamba	MiM	MambaHSI	MambaLG	IGroupSS-Mamba	Ours
1	81.99±3.56	90.14±2.84	90.49±1.89	84.02±2.25	96.94±1.92	82.08±2.92	96.67±1.63	87.80±6.77	93.38±1.01	89.28±2.19	91.33±0.79	**98.57±0.46**
2	95.51±1.27	97.17±1.59	95.58±2.54	78.24±2.33	85.90±6.97	88.65±3.81	98.74±0.55	96.79±0.97	95.90±2.02	88.07±8.13	95.30±0.56	**98.78±0.40**
3	73.07±2.71	78.23±6.38	70.90±3.60	77.73±0.30	90.96±4.83	66.60±2.00	92.66±3.38	80.75±2.35	94.82±4.11	87.25±3.02	88.84±0.20	**95.83±2.39**
4	95.36±1.10	92.29±2.57	90.16±6.98	89.97±1.27	91.86±4.07	96.26±1.96	96.00±0.88	93.17±0.83	92.18±1.26	**97.57±0.58**	97.22±0.03	94.32±2.62
5	99.44±0.38	98.26±1.03	99.78±0.27	100.0±0.00	99.55±0.25	99.66±0.42	99.48±0.36	98.45±0.64	99.98±0.03	100.0±0.00	**100.0±0.00**	99.57±0.25
6	72.59±11.8	80.90±7.77	66.12±6.29	93.86±2.83	93.19±5.30	74.35±9.82	98.43±0.86	86.88±2.83	98.55±0.92	97.93±0.60	98.20±0.55	**98.94±0.89**
7	81.53±8.61	92.88±6.94	97.19±2.31	98.86±0.28	87.67±21.7	83.10±2.71	**99.02±1.61**	92.05±5.61	97.30±2.00	94.62±2.75	96.69±0.86	98.67±1.61
8	95.34±1.14	97.84±1.09	98.04±0.95	94.18±0.85	95.43±1.72	96.06±0.79	98.55±0.80	95.60±2.06	94.52±3.35	**98.96±0.86**	98.22±0.95	98.40±0.63
9	92.03±1.94	95.72±1.33	98.72±1.21	98.99±0.32	98.06±0.91	92.86±1.05	97.76±0.69	92.20±3.12	**99.45±0.31**	97.01±0.77	96.23±0.02	97.48±1.25
OA(%)	89.41±1.59	92.94±1.05	90.32±1.97	84.63±1.10	90.60±2.87	86.41±0.56	97.90±0.31	92.99±1.51	95.63±0.96	91.62±3.92	95.28±0.35	**98.27±0.27**
AA(%)	87.43±0.90	91.49±1.19	89.67±1.57	90.65±0.40	93.29±3.38	86.62±1.15	97.48±0.40	91.52±1.67	96.23±0.74	94.52±1.37	95.78±0.03	**97.84±0.46**
$k\times100$	85.65±2.22	90.43±1.43	86.82±2.65	80.10±1.37	87.66±3.65	81.87±0.69	97.16±0.41	90.54±2.01	94.75±2.78	89.06±4.95	93.69±0.46	**97.66±0.37**

more accurate complementary fusion, reducing misclassifications in uncommon conditions.

The results of various models counted on HUS2013 are shown in Table VII. This dataset is large in pixel volume and intricate in land cover distribution, which increases classification difficulty. Consequently, there is a substantial variation in the performance across different models. Even so, SpiralMamba still delivers the optimal performance. For example, for the Kappa coefficient, our enhancements are 7.68 (over SpectralFormer), 1.89 (over MorphFormer), 6.98 (over SQSFormer), 4.33 (over GraphGST), 3.28 (over DBCTNet), 5.06 (over HSIMamba), 1.11 (over S^2Mamba), 5.16 (over MiM), 0.87 (over MambaHSI), 4.64(over MambaLG) and 3.13 (over IGroupSS-Mamba). These encouraging results demonstrate the advantages of SpiralMamba and capability to handle complex data. However, we cannot ignore a notable investigation, that is, SpiralMamba does not achieve the highest accuracy in some categories. Especially for class 1 ("Healthy Grass") and class 3 ("Synthetic Grass"), the gaps between the proposed model and the optimal methods are distinct. The main reason behind this is the insufficient local feature exploration of SpiralMamba in spatial contexts. Particularly, the classes mentioned above are spatially adjacent and similar to visual. Therefore, the difficulty of recognizing them is inherently high. To successfully classify them, more local details should be captured. However, our SpiralMamba pays more attention to extracting spatial global context information. The local spatial clues do not receive the attention they deserve, resulting in unsatisfactory results in these categories. We plan to improve this aspect in future work to achieve better performance across more categories.

The results of all methods on HUS2018 are summarized in Table VIII. Our SpiralMamba achieves the best performance across three metrics, reaching 84.03% (OA), 89.67% (AA), and 79.69 (Kappa), respectively. Compared to Transformer-based methods, our approach outperforms the best-performing DBCTNet by 2.55% and the lowest-performing SpectralFormer by 15.29% in OA. These improvements stem from two factors. First, SpiralMamba adopts the sequence model Mamba, which is better suited for directly capturing spectral sequences, enhancing efficiency and performance. Second, our dual-branch fusion architecture can comprehensively extract spatial–spectral information and achieve more accurate fusion. Compared to Mamba-based methods, SpiralMamba surpasses the best-performing MambaHSI by 2.4% and the lowest-performing HSIMamba by 12.27% in OA. This success can be attributed to the spiral scan and the MLC module, which effectively convey spatial information and perceive local details within spectral tokens. Together, these components enable more accurate and comprehensive modeling of spatial and spectral features, leading to superior classification results.

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TABLE VII

PERFORMANCE OF DIFFERENT METHODS COUNTED ON HUS2013, INCLUDING THE RESULTS OF EACH CLASS AND THE OA, AA, AND KAPPA COEFFICIENT VALUES. THE BEST RESULTS ARE BOLDED (MEAN ± VARIANCE)

Class No.	Transformer-based Methods					Mamba-based Methods						
	SpectralFormer	MorphFormer	SQSFormer	GraphGST	DBCTNet	HSIMamba	S^2Mamba	MiM	MambaHSI	MambaLG	IGroupSS-Mamba	Ours
1	83.55±0.99	83.13±2.23	86.23±3.13	88.89±3.06	82.37±0.99	**97.49±2.80**	83.36±0.63	82.91±2.31	94.94±3.65	81.65±0.65	82.00±0.05	83.04±1.66
2	98.89±0.64	98.16±1.57	94.02±2.95	99.72±0.18	98.48±1.34	98.93±0.50	99.36±0.77	98.23±1.56	94.48±5.47	98.62±1.31	99.76±0.24	**99.94±0.11**
3	91.21±1.94	95.60±3.85	**99.92±0.10**	96.04±0.33	99.45±0.38	93.90±1.83	99.05±0.46	99.60±0.00	99.55±0.64	97.26±0.97	96.83±1.19	94.50±2.36
4	98.62±0.62	97.37±0.59	94.02±1.42	91.38±1.23	96.34±1.43	93.43±3.33	**99.62±0.30**	92.01±4.16	95.77±2.09	99.11±1.14	99.52±0.29	99.09±0.46
5	99.32±0.69	98.88±0.57	100.0±0.00	100.0±0.00	99.85±0.14	98.82±0.11	100.0±0.00	100.0±0.00	99.34±0.97	100.0±0.00	**100.0±0.00**	99.96±0.08
6	93.99±0.84	97.48±2.06	99.30±1.40	**100.0±0.00**	95.94±0.28	99.44±0.52	97.34±1.90	95.80±0.00	97.49±2.62	95.66±1.56	95.10±0.69	97.62±2.92
7	88.82±2.43	91.64±2.25	83.21±2.47	79.91±1.85	**93.77±3.00**	88.62±0.85	90.37±1.57	83.47±2.85	92.46±4.03	82.89±4.71	87.69±2.90	91.08±2.34
8	76.32±5.82	87.79±3.22	79.81±3.76	90.48±0.44	85.20±5.09	84.25±1.95	90.47±1.12	84.46±8.14	67.91±4.07	81.84±5.32	84.47±1.28	**91.45±2.53**
9	77.64±2.65	89.84±0.69	80.64±5.58	76.26±1.42	80.98±4.63	80.09±2.95	84.15±2.19	84.71±1.57	85.04±5.53	86.29±2.17	86.92±0.99	**90.25±1.25**
10	81.66±7.39	86.00±2.32	64.69±1.90	89.25±1.86	81.49±13.5	83.24±3.63	84.92±2.88	72.64±1.88	92.94±4.72	76.04±7.99	81.13±6.61	**93.22±3.41**
11	77.57±6.57	96.34±1.32	90.36±2.88	87.89±4.11	94.63±5.21	81.12±1.11	98.46±0.79	95.10±4.09	93.72±3.84	93.76±3.89	91.98±3.18	**98.80±0.93**
12	82.40±3.41	91.24±2.65	91.09±0.53	90.22±3.24	89.32±2.49	80.40±2.62	88.36±3.82	91.82±0.86	87.32±4.31	88.09±2.81	90.73±0.14	**92.10±1.30**
13	71.86±5.67	90.04±3.33	80.07±3.62	81.68±0.93	90.60±1.79	85.96±1.11	**91.09±1.34**	75.23±8.58	84.80±6.69	89.54±1.39	91.05±0.17	89.19±3.11
14	99.35±0.49	98.14±1.75	100.0±0.00	100.0±0.00	99.11±1.78	98.06±0.90	100.0±0.00	100.0±0.00	100.0±0..00	98.70±1.82	100.0±0.00	**100.0±0.00**
15	97.97±1.75	95.69±4.91	100.0±0.00	100.0±0.00	97.76±3.08	99.79±0.33	100.0±0.00	99.92±0.17	96.25±3.53	99.92±0.10	99.79±0.02	**100.0±0.00**
OA(%)	87.15±0.52	92.49±0.32	87.80±0.76	90.24±0.75	91.20±1.72	89.57±0.34	93.20±0.26	89.47±0.78	90.77±0.61	89.93±1.26	91.33±0.29	**94.23±0.36**
AA(%)	87.94±0.72	93.16±0.55	89.56±0.81	91.45±0.61	92.35±1.34	90.90±0.26	94.13±0.23	90.39±0.92	92.13±0.57	91.29±1.02	92.47±0.14	**94.69±0.39**
$k\times100$	86.05±0.56	91.84±0.35	86.75±0.83	89.40±0.82	90.45±1.86	88.67±0.37	92.62±0.28	88.57±0.84	92.86±1.75	89.09±1.36	90.60±0.31	**93.73±0.39**

TABLE VIII

PERFORMANCE OF DIFFERENT METHODS COUNTED ON HUS2018, INCLUDING THE RESULTS OF EACH CLASS AND THE OA, AA, AND KAPPA COEFFICIENT VALUES. THE BEST RESULTS ARE BOLDED (MEAN ± VARIANCE)

Class No.	Transformer-based Methods					Mamba-based Methods						
	SpectralFormer	MorphFormer	SQSFormer	GraphGST	DBCTNet	HSIMamba	S^2Mamba	MiM	MambaHSI	MambaLG	IGroupSS-Mamba	Ours
1	91.62±0.93	90.89±3.10	90.47±1.08	88.83±2.21	82.28±4.51	82.67±4.78	91.41±2.15	91.05±1.08	87.30±1.01	88.37±2.62	87.57±2.53	**92.22±3.96**
2	84.52±0.40	86.00±2.55	83.10±1.10	85.25±1.49	**90.22±3.23**	84.71±2.60	85.87±0.34	75.82±1.01	88.97±2.39	85.36±2.95	83.71±2.45	88.14±2.70
3	99.57±0.09	99.86±1.13	100.0±0.00	99.93±0.08	99.94±0.08	99.49±0.28	99.97±0.07	99.42±0.12	99.91±0.09	100.0±0.00	99.79±0.13	**100.0±0.00**
4	94.90±0.54	96.17±0.52	97.33±0.67	93.41±1.85	94.53±2.79	93.51±2.33	96.87±1.47	93.41±1.86	96.97±0.09	96.95±0.42	97.23±0.78	**97.48±0.40**
5	88.95±0.81	89.03±2.60	**94.59±1.44**	89.68±2.14	87.01±5.02	85.78±4.26	87.67±1.95	77.48±1.97	92.49±0.60	93.98±0.97	93.42±1.37	90.10±2.83
6	97.83±0.91	99.45±0.44	99.44±0.41	99.79±0.11	99.74±0.21	98.60±0.17	99.27±0.84	99.64±0.29	99.68±0.05	99.79±0.06	99.76±0.04	**99.81±0.26**
7	99.70±0.30	100.0±0.00	100.0±0.00	99.88±0.24	99.60±0.28	99.60±0.57	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	**100.0±0.00**
8	75.41±2.92	83.71±1.82	87.80±1.41	87.76±0.80	83.14±5.89	74.86±4.33	86.28±3.06	85.51±0.36	86.58±1.77	86.97±2.05	**88.89±1.55**	86.57±3.42
9	68.00±4.00	82.54±2.91	87.19±1.23	77.30±0.90	86.30±1.78	74.13±0.46	85.89±2.29	85.58±0.42	87.19±0.53	87.70±0.97	86.83±1.00	**88.01±1.94**
10	48.57±7.52	54.92±2.93	57.89±2.42	47.43±4.51	54.87±2.86	47.84±3.47	57.04±1.53	54.19±1.14	52.13±3.57	57.76±1.87	52.58±8.44	**64.48±5.18**
11	50.54±1.45	61.49±3.42	55.01±1.67	54.11±3.64	**69.77±5.33**	50.64±3.42	57.06±3.36	61.99±2.46	58.24±4.55	53.89±2.71	58.70±4.45	63.32±4.55
12	63.89±2.05	65.30±7.86	73.77±0.61	63.96±1.80	72.31±10.4	66.27±1.42	72.79±5.14	78.17±5.61	75.49±1.80	69.44±1.95	**85.11±4.30**	67.77±3.66
13	48.81±2.18	64.28±4.72	58.77±4.86	70.16±3.58	68.32±2.52	55.69±4.33	61.10±3.43	73.81±1.92	69.87±1.75	65.32±1.49	69.28±3.38	**74.49±2.50**
14	87.11±0.07	93.57±2.62	93.71±4.15	**97.03±1.59**	89.38±1.95	89.47±0.49	97.02±1.90	92.84±3.12	91.63±4.62	92.61±3.84	93.46±3.63	94.18±2.60
15	97.75±1.40	98.76±0.95	99.08±0.48	98.62±1.29	93.01±7.62	95.62±0.52	98.05±0.64	99.62±0.07	98.41±0.52	99.18±3.84	99.24±0.31	**99.25±0.27**
16	82.75±2.45	88.48±3.49	91.61±2.14	88.81±1.10	86.72±5.55	85.39±2.17	88.20±2.92	89.59±4.67	88.79±1.59	91.42±0.76	92.00±2.98	**92.30±2.82**
17	100.00±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	100.0±0.00	**100.0±0.00**
18	87.97±1.09	95.74±1.09	94.09±1.08	96.29±0.45	93.36±3.23	88.03±1.47	95.85±1.33	95.48±1.23	96.28±0.50	92.42±1.68	96.35±0.41	**96.37±1.73**
19	95.34±1.79	98.52±1.33	96.53±3.09	98.90±0.52	98.98±0.33	93.72±0.84	98.88±0.85	97.65±0.75	98.57±0.29	98.19±0.95	99.07±0.52	**99.11±0.57**
20	95.42±0.49	97.52±0.83	99.39±0.35	99.58±0.50	99.26±0.26	96.79±0.13	98.41±0.79	99.13±0.27	99.68±0.04	99.75±0.23	99.73±0.13	**99.77±0.20**
OA(%)	68.74±2.00	79.06±1.11	80.77±0.34	76.38±0.73	81.48±0.75	71.76±0.39	80.45±1.12	80.59±0.39	81.63±0.33	81.49±0.48	81.51±1.04	**84.03±0.97**
AA(%)	83.18±0.43	87.31±0..50	88.00±0.55	86.89±1.15	87.44±1.79	83.14±0.06	87.88±0.42	87.48±0.33	88.49±0.07	87.96±0.19	89.22±0.28	**89.67±0.25**
$k\times100$	62.14±2.05	73.73±1.24	75.64±0.48	70.08±1.44	76.52±0.92	65.26±0.45	75.27±1.32	75.48±0.45	76.67±0.35	76.53±0.55	76.62±1.24	**79.69±1.13**

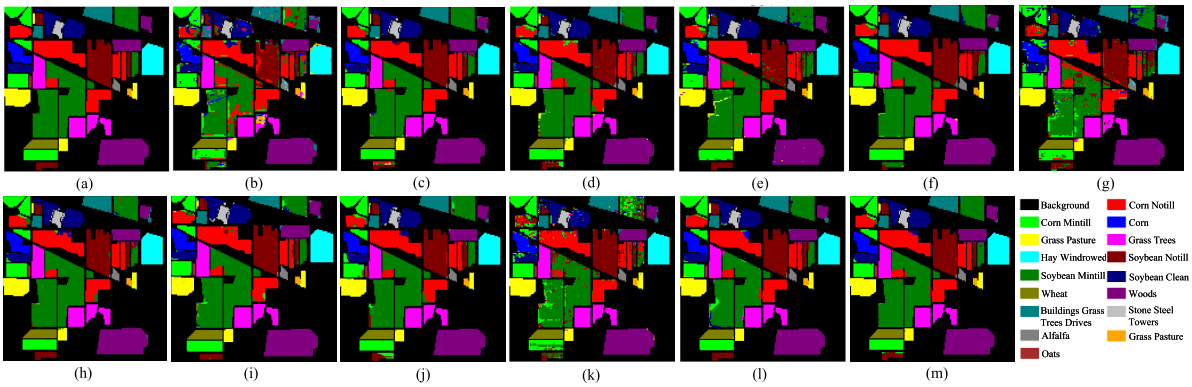


Fig. 8. Ground truth and classification maps of different methods on IP. (a) Ground truth. (b) SpectralFormer. (c) MorphFormer. (d) SQSFormer. (e) GraphGST. (f) DBCTNet. (g) HSIMamba. (h) S^2Mamba. (i) MiM. (j) MambaHSI. (k) MambaLG. (l) IGroupSS-Mamba. (m) Ours.

2) Comparison of Visual Results: To further study SpiralMamba, we visualize its classification maps for the four datasets in Figs. 8–11. The results of the compared methods are also exhibited for reference. For IP (see Fig. 8), the classification maps of SpectralFormer, HSIMamba, GraphGST, and MambaLG differ significantly from the ground truth, showing dense artifacts of misclassification. Although the visual results of MorphFormer, SQSFormer, DBCTNet, S^2Mamba,

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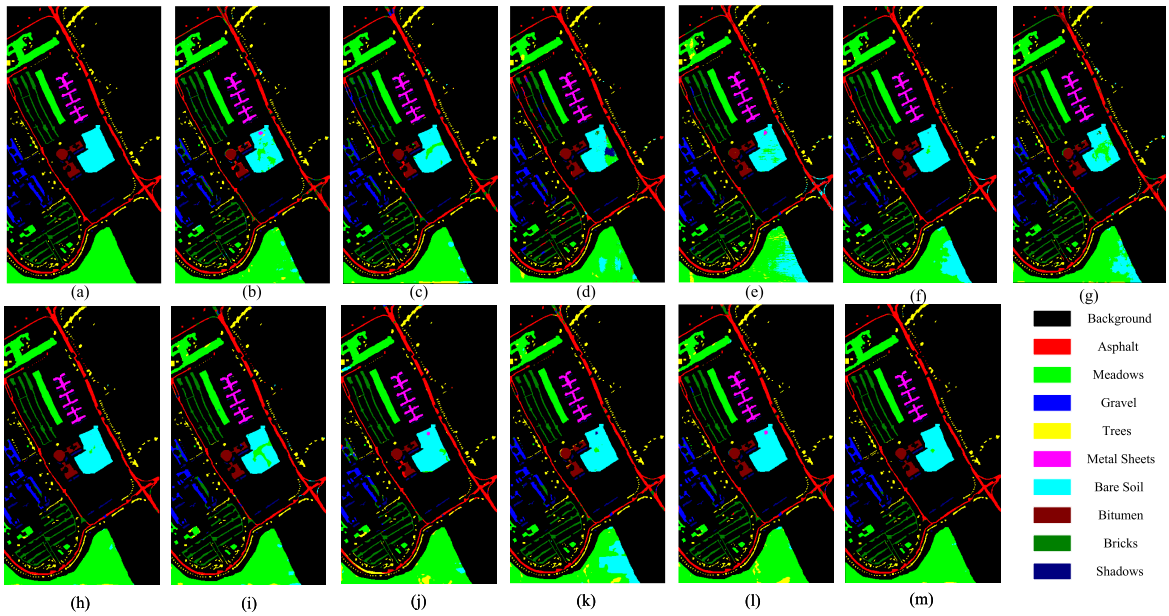


Fig. 9. Ground truth and classification maps of different methods on PU. (a) Ground truth. (b) SpectralFormer. (c) MorphFormer. (d) SQSFormer. (e) GraphGST. (f) DBCTNet. (g) HSIMamba. (h) S^2Mamba. (i) MiM. (j) MambaHSI. (k) MambaLG. (l) IGroupSS-Mamba. (m) Ours.

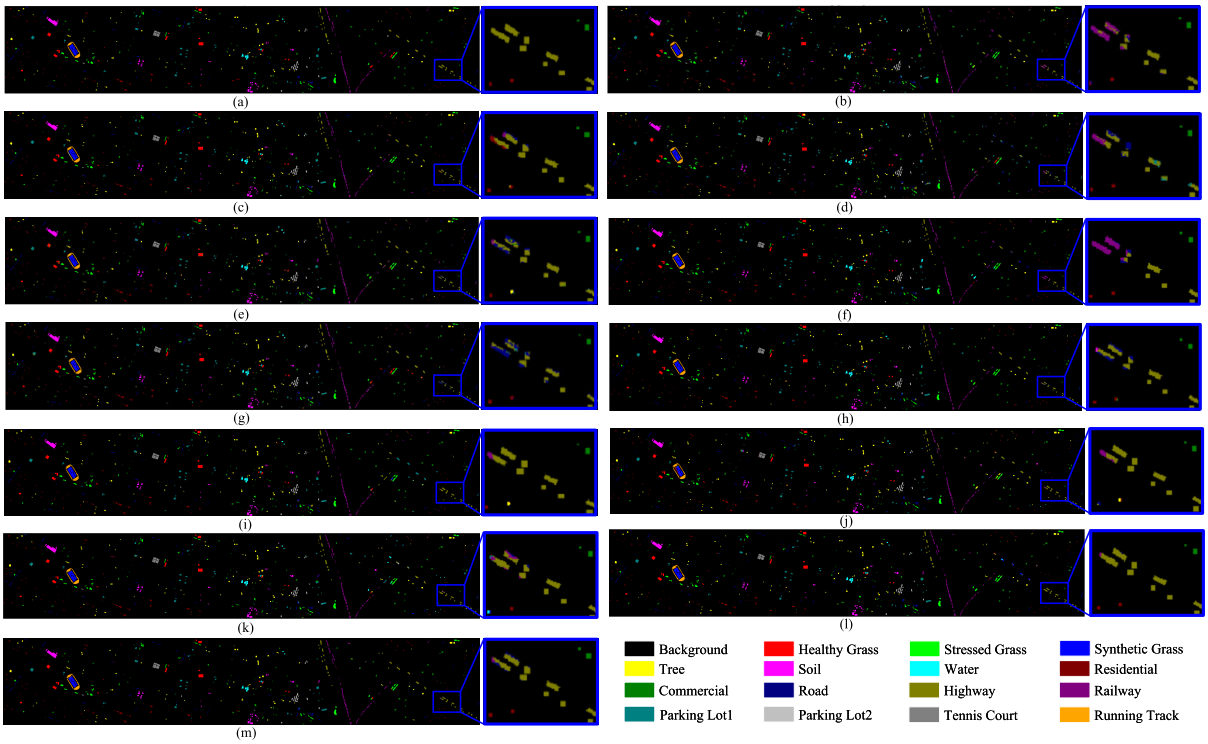


Fig. 10. Ground truth and classification maps of different methods on HUS2013. (a) Ground truth. (b) SpectralFormer. (c) MorphFormer. (d) SQSFormer. (e) GraphGST. (f) DBCTNet. (g) HSIMamba. (h) S^2Mamba. (i) MiM. (j) MambaHSI. (k) MambaLG. (l) IGroupSS-Mamba. (m) Ours.

MiM, MambaHSI, and IGroupSS-Mamba are better, they still exhibit misclassification errors at the land cover boundaries. In contrast, our SpiralMamba shows the highest similarity with the ground truth, achieving relatively accurate classifications in smooth and boundary regions. For PU (see Fig. 9), all compared models perform weakly in distinguishing the "Bare Soil" (light blue in the center) and "Meadows" (light green at the bottom). The distinct errors can be found in SQSFormer, GraphGST, DBCTNet, HSIMamba, and MambaLG's maps, where the "Bare Soil" is misclassified into "Meadows," "Shadows," "Metal Sheets," or "Bricks." SpectralFormer, MorphFormer, S^2Mamba, MiM, MambaHSI, and IGroupSS-Mamba's maps are better but fail to reach satisfactory results. In comparison, our SpiralMamba exhibits the fewest misclassifications and presents the most similar predictions to the ground truth. For HUS2013 (see Fig. 10), SpectralFormer, SQSFormer, DBCTNet, and HSIMamba show the powerless performance. In the enlarged detail sections, these models

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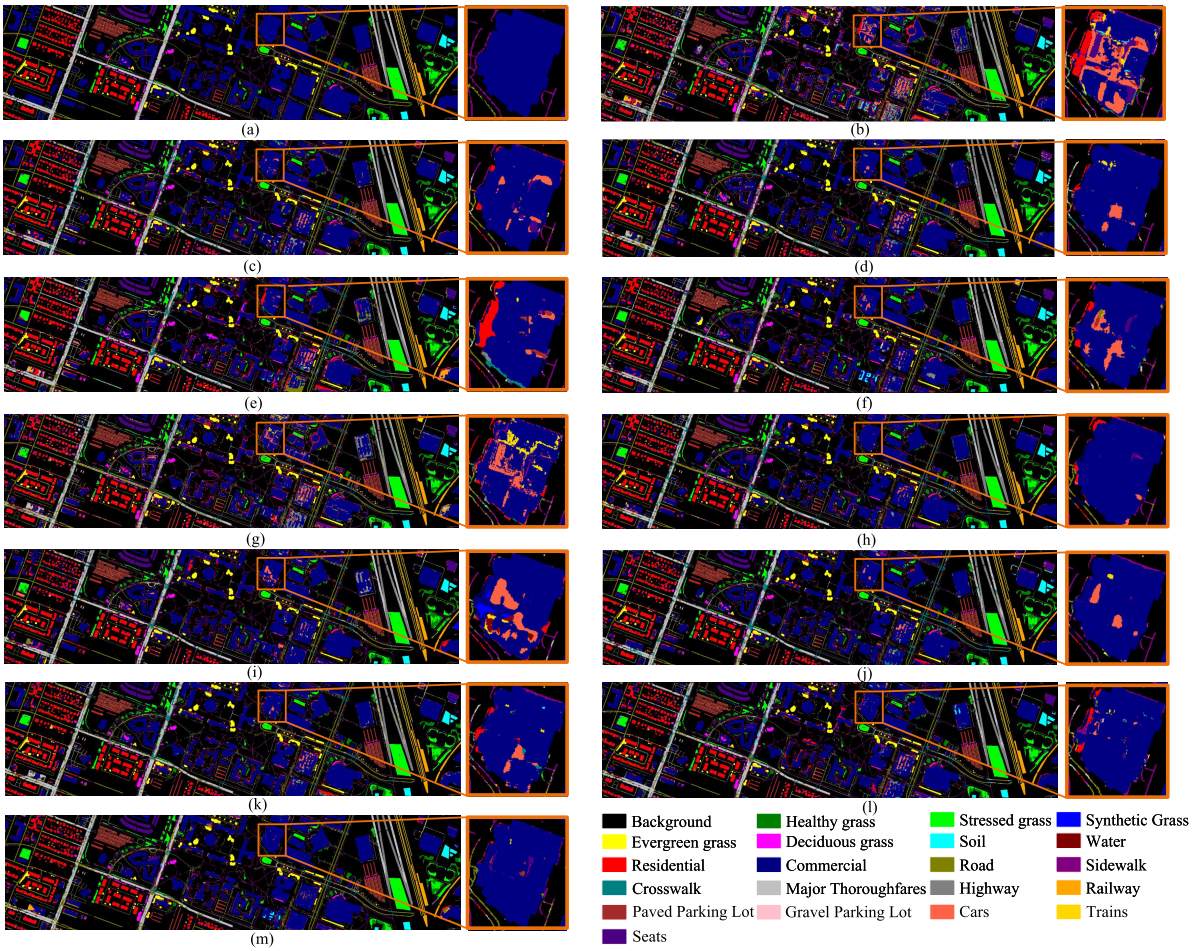


Fig. 11. Ground truth and classification maps of different methods on HUS2018. (a) Ground truth. (b) SpectralFormer. (c) MorphFormer. (d) SQSFormer. (e) GraphGST. (f) DBCTNet. (g) HSIMamba. (h) S^{2}Mamba. (i) MiM. (j) MambaHSI. (k) MambaLG. (l) IGroupSS-Mamba. (m) Ours.

misclassify "Highway" (olive green) as "Soil" (pink) or "Road" (dark blue). MorphFormer, GraphGST, S^{2}Mamba, MiM MambaHSI, MambaLG, and IGroupSS-Mamba perform slightly better but still struggle with edge areas. In contrast, our SpiralMamba produces the best prediction results. For HUS2018 (see Fig. 11), methods, such as SpectralFormer, GraphGST, DBCTNet, HSIMamba, and MiM, exhibit suboptimal classification maps. Particularly in zoomed-in regions, flat "Commercial" areas are misclassified as "Residential," "Cars," or "Trains." In contrast, MorphFormer, SQSFormer, S^{2}Mamba, MambaHSI, MambaLG, and IGroupSS-Mamba show fewer misclassifications. SpiralMamba produces a classification map that is most similar to the ground truth. Although minor misclassifications remain in the central zoomed-in region, these are limited to similar categories like "Residential" rather than dissimilar ones like "Cars." This is because our method leverages Mamba's global modeling capability to accurately interpret flat and concentrated regions (e.g., "commercial") from a global perspective, minimizing the impact of local noise and yielding a satisfactory classification map.

Overall, in challenging scenarios, such as boundary regions and confusable classes, most models struggle to make accurate distinctions, which negatively impacts overall classification performance. However, our model maintains a relatively superior classification ability, achieving predictions similar to the ground truth. This is attributed not only to the advanced spatial modeling capability of SpaME and the powerful spectral feature extraction ability of SpeME but also to the accurate and effective spatial–spectral fusion by SSCFM. The synergy of these components enables SpiralMamba to deliver satisfactory classification results.

D. Ablation Studies

To validate the contributions of different components of SpiralMamba, a series of ablation experiments are conducted under the same experimental environment and settings as described in Section IV-B1. These experiments include overall ablation, scanning strategy ablation, and fusion module ablation.

1) Overall Ablation: We performed an incremental ablation for the proposed spiral scan, MLC, and SSCFM by either using or excluding each component. When spiral scan and SSCFM are not used, the simplest operations (flatten and addition) are applied as replacements. Table IX shows the detailed experimental results. It can be observed that there is the same pattern in four datasets. The baseline without these components yields the lowest performance. Using spiral scan, MLC, or SSCFM individually improves results, combining any two components

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TABLE IX

OVERALL ABLATION EXPERIMENTAL RESULTS OF EACH COMPONENT ON IP, PU, HUS2013, AND HUS2018. √ MEANS THE COMPONENT IS USED, AND BLANK MEANS IT IS NOT USED. THE BEST RESULT IS BOLDED (MEAN ± VARIANCE)

Experiment No.	Spiral Scan	MLC	SSCFM	Indian Pines			Pavia University			Houston 2013			Houston 2018		
				OA(%)	AA(%)	$k\times100$	OA(%)	AA(%)	$k\times100$	OA(%)	AA(%)	$k\times100$	OA(%)	AA(%)	$k\times100$
1				95.33±0.65	97.46±0.34	94.64±0.74	96.77±1.02	96.96±0.45	95.66±1.35	93.45±0.06	93.99±0.29	92.89±0.06	79.14±0.76	87.11±0.58	73.73±0.85
2	√			95.68±0.71	97.66±0.20	95.05±0.81	97.41±0.33	97.18±0.50	96.50±0.45	93.58±0.19	94.08±0.22	93.03±0.20	80.01±1.41	87.84±0.87	74.81±1.60
3		√		97.30±0.31	98.44±0.26	96.89±0.35	97.59±0.82	97.12±0.66	96.74±1.12	93.68±0.52	94.09±0.57	93.14±0.57	79.96±0.45	88.37±0.35	74.81±0.53
4			√	95.99±0.66	97.78±0.27	95.40±0.75	97.18±0.18	96.89±0.75	96.19±1.06	93.71±0.39	94.27±0.29	93.18±0.42	80.02±1.20	87.72±0.34	74.82±1.35
5	√	√		97.31±0.21	98.51±0.13	96.92±0.24	97.81±0.21	97.68±0.30	97.05±0.29	93.91±0.15	94.34±0.12	93.39±0.17	83.03±0.91	88.88±0.62	78.45±1.08
6	√		√	96.26±0.33	97.87±0.35	95.71±0.38	98.12±0.37	97.45±0.77	97.47±0.50	94.11±0.19	94.19±0.14	93.61±0.20	81.44±1.27	88.04±0.39	76.31±1.44
7		√	√	97.56±0.12	98.66±0.11	97.19±0.14	98.03±0.19	97.57±0.45	97.34±0.26	93.72±0.38	94.28±0.29	93.19±0.42	82.73±0.43	88.50±0.15	77.11±0.45
8	√	√	√	**97.76±0.27**	**98.76±0.19**	**97.43±0.31**	**98.27±0.27**	**97.84±0.46**	**97.66±0.37**	**94.23±0.36**	**94.69±0.39**	**93.73±0.39**	**84.03±0.97**	**89.67±0.25**	**79.69±1.13**

. √ means the component is used, and blank means it is not used. The best result is **bolded**. (mean±variance)

TABLE X

EXPERIMENTAL RESULTS OF DIFFERENT SCANNING STRATEGIES ON IP, PU, HUS2013, AND HUS2018. THE BEST RESULT IS BOLDED (MEAN ± VARIANCE)

Scan Strategy	Indian Pines			Pavia University			Houston 2013			Houston 2018		
	OA(%)	AA(%)	$k\times100$	OA(%)	AA(%)	$k\times100$	OA(%)	AA(%)	$k\times100$	OA(%)	AA(%)	$k\times100$
Flatten	97.56±0.12	98.66±0.11	97.19±0.14	98.03±0.19	97.57±0.45	97.34±0.26	93.72±0.38	94.28±0.29	93.19±0.42	82.73±0.43	88.50±0.15	77.11±0.45
Random Scan	96.95±0.14	98.41±0.02	96.50±0.15	97.65±0.33	97.09±0.72	96.83±0.44	93.55±0.35	93.24±0.22	92.99±0.38	81.13±0.40	85.82±0.70	76.00±0.50
Cross Scan	97.40±0.16	98.55±0.08	97.02±0.19	97.90±0.11	97.29±0.27	97.16±0.15	93.65±0.34	93.93±0.25	93.11±0.37	82.04±0.75	86.38±0.86	77.01±0.53
Spiral Scan	**97.76±0.27**	**98.76±0.19**	**97.43±0.31**	**98.27±0.27**	**97.84±0.46**	**97.66±0.37**	**94.23±0.36**	**94.69±0.39**	**93.73±0.39**	**84.03±0.97**	**89.67±0.25**	**79.69±1.13**

. The best result is **bolded**. (mean±variance)

TABLE XI

EXPERIMENTAL RESULTS OF SPIRAL SCAN WITH DIFFERENT SCANNING DIRECTIONS. THE BEST RESULT IS BOLDED

Spiral Scan	Direction	Indian Pines			Pavia University			Houston 2013			Houston 2018		
		OA(%)	AA(%)	$k\times100$	OA(%)	AA(%)	$k\times100$	OA(%)	AA(%)	$k\times100$	OA(%)	AA(%)	$k\times100$
Single Spiral	1	97.57±0.20	98.55±0.16	97.21±0.24	98.08±0.42	97.54±0.28	97.41±0.56	93.75±0.35	94.18±0.30	93.22±0.38	83.15±1.14	88.72±0.71	78.60±1.41
Positive Spiral	4	97.63±0.13	98.69±0.09	97.28±0.15	98.18±0.06	97.61±0.12	97.54±0.08	93.96±0.36	94.71±0.26	93.44±0.39	83.69±0.61	89.11±0.90	79.31±0.68
Reverse Spiral	4	97.58±0.09	98.60±0.09	97.22±0.10	98.11±0.29	97.71±0.21	97.44±0.39	93.77±0.24	94.20±0.19	93.23±0.26	83.29±0.51	88.81±1.98	78.76±0.61
All Spiral	8	**97.76±0.27**	**98.76±0.19**	**97.43±0.31**	**98.27±0.27**	**97.84±0.46**	**97.66±0.37**	**94.23±0.36**	**94.69±0.39**	**93.73±0.39**	**84.03±0.97**	**89.67±0.25**	**79.69±1.13**

further enhances performance, and employing all three (the complete SpiralMamba) achieves optimal performance. This is because spiral scan is well-suited for HSIs, accurately transforming spatial information into the 1-D sequence, aiding Mamba in understanding spatial context. Additionally, MLC's fine-grained local modeling within spectral tokens and SSCFM's enhanced spatial–spectral relationship modeling and refined fusion contribute to superior performance.

2) Scanning Strategy Ablation: To further validate the effectiveness of the proposed spiral scan, we only change the spatial scanning strategy based on the complete SpiralMamba. Expect the flatten operation, the comparisons with the cross scan [42] and random scan are also conducted. The experimental results are shown in Table X. It is obvious that SpiralMamba with spiral scan achieves the best performance across four datasets. This is because flatten has a single scanning direction and suffers from information discontinuities at the edge pixels. Although cross scan has more directions, its performance cannot even exceed flatten, which means that it may introduce more spatial discontinuities, causing nonadjacent pixels to become adjacent in the sequence. Random scan is more flexible, but its randomness disrupts the spatial locations, interfering with the model's learning. In contrast, our spiral scan covers all scanning directions and ensures the continuity of spatial information. More importantly, it takes into account the characteristics of HSI patches expanding from the center pixel, emphasizes the importance of the center pixel, and preserves the relative positions of neighboring pixels to the center pixel. This allows Mamba to understand the spatial layout better, improving HSI classification performance.

To study the impacts of the spiral scan direction on the model's performance, we conducted experiments with four types of spiral scans: single, positive, reverse, and all spiral scan. The single spiral scan follows one direction, starting from the top-left pixel and ending at the central pixel. The positive and reverse spiral scans involve four directional paths, starting from the four corners and the central pixel. All spiral scan represents the complete SpiralMamba with eight directional paths. The results counted on four HSIs are summarized in Table XI. By observing it, we find that all spiral scan produced the best performance, followed by the positive spiral scan, then the reverse spiral scan, and the single spiral scan yielding the lowest performance. The reasons behind these findings are summarized as follows. First, the single-direction scan has only one fixed traversal path, which limits Mamba from fully understanding spatial information. Second, the four-direction positive/reverse spiral scan increases the number of paths, providing Mamba with more spatial awareness and leading to some performance improvement. Third, when combining positive and reverse spiral scans, all spiral scan grants SpiralMamba the ability to capture spatial context bidirectionally. This accounts for the encouraging classification results.

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TABLE XII

COMPARISON OF THE TIME COMPLEXITY AND PERFORMANCE OF SPIRAL SCAN AND FLATTEN ON SPIRALMAMBA. THE BEST RESULT IS BOLDED

Datasets	Scan Strategy	Params (*MB*)	Memory (*MB*)	Training (*s/epoch*)	Inference (*s*)	OA (%)	AA (%)	Kappa (×*100*)
Indian Pines	Flatten	**1.63**	**1476**	**0.40**	**1.80**	97.56±0.12	98.66±0.11	97.19±0.14
	Spiral Scan	1.97	1742	0.70	2.10	**97.76±0.27**	**98.76±0.19**	**97.43±0.31**
Pavia University	Flatten	**0.52**	**1018**	**1.45**	**3.76**	98.03±0.19	97.57±0.45	97.34±0.26
	Spiral Scan	0.55	1042	2.02	4.29	**98.27±0.27**	**97.84±0.46**	**97.66±0.37**
Houston 2013	Flatten	**1.03**	**1008**	**0.95**	**1.34**	93.72±0.38	94.28±0.29	93.19±0.42
	Spiral Scan	1.05	1026	1.34	1.72	**94.23±0.36**	**94.69±0.39**	**93.73±0.39**
Houston 2018	Flatten	**0.14**	**2206**	**0.70**	**21.08**	82.73±0.43	88.50±0.15	77.11±0.45
	Spiral Scan	0.15	3928	0.93	25.63	**84.03±0.97**	**89.67±0.25**	**79.69±1.13**

TABLE XIII

EXPERIMENTAL RESULTS RELATED TO MLC ON IP, PU, HUS2013, AND HUS2018. THE BEST RESULT IS BOLDED (MEAN ± VARIANCE)

Method	Indian Pines			Pavia University			Houston 2013			Houston 2018		
	OA(%)	AA(%)	k×100	OA(%)	AA(%)	k×100	OA(%)	AA(%)	k×100	OA(%)	AA(%)	k×100
with None	96.26±0.33	97.87±0.35	95.71±0.38	98.12±0.37	97.45±0.77	97.47±0.50	94.11±0.19	94.19±0.14	93.61±0.20	81.44±1.27	88.04±0.39	76.31±1.44
with MLP	96.56±0.23	98.15±0.14	96.06±0.27	98.13±0.37	97.82±0.38	97.48±0.48	94.16±0.44	94.17±0.37	93.66±0.47	81.56±0.53	88.47±0.92	76.59±1.28
with MLC	**97.76±0.27**	**98.76±0.19**	**97.43±0.31**	**98.27±0.27**	**97.84±0.46**	**97.66±0.37**	**94.23±0.36**	**94.69±0.39**	**93.73±0.39**	**84.03±0.97**	**89.67±0.25**	**79.69±1.13**

TABLE XIV

EXPERIMENTAL RESULTS OF DIFFERENT FUSION METHODS ON IP, PU, HUS2013, AND HUS2018. THE BEST RESULT IS BOLDED (MEAN ± VARIANCE)

Fusion Method	Indian Pines			Pavia University			Houston 2013			Houston 2018		
	OA(%)	AA(%)	k×100	OA(%)	AA(%)	k×100	OA(%)	AA(%)	k×100	OA(%)	AA(%)	k×100
SpaME only	97.30±0.31	98.44±0.26	96.89±0.35	97.55±0.21	97.76±0.32	96.71±0.28	93.70±0.51	94.58±0.43	93.16±0.56	80.64±0.86	86.48±0.73	76.45±0.98
SpeME only	97.44±0.38	98.64±0.13	97.06±0.44	97.19±0.40	97.05±0.21	96.22±0.52	93.30±0.28	94.17±0.24	92.73±0.31	79.86±1.04	85.95±0.94	76.10±1.02
SpaME+SpeME(add)	97.31±0.21	98.51±0.13	96.92±0.24	97.81±0.21	97.68±0.30	97.05±0.29	93.91±0.15	94.34±0.12	93.39±0.17	83.03±0.91	88.88±0.62	78.45±1.08
SpaME+SpeME(cat)	97.31±0.29	98.46±0.13	96.91±0.33	97.91±0.25	97.43±0.25	97.18±0.33	94.12±0.39	94.63±0.48	93.62±0.42	83.59±0.72	88.91±0.65	79.13±0.88
SpaME+SpeME(SSCFM)	**97.76±0.27**	**98.76±0.19**	**97.43±0.31**	**98.27±0.27**	**97.84±0.46**	**97.66±0.37**	**94.23±0.36**	**94.69±0.39**	**93.73±0.39**	**84.03±0.97**	**89.67±0.25**	**79.69±1.13**

. The best result is **bolded**. (mean±variance)

In addition, to study if the proposed spiral scanning technique increases the computational complexity or not, we replace it with flatten and count the new model's parameters (Params), memory usage in the training phase (Memory), average training times per epoch (Training), and the inference time costs (Inference). The results counted on different HSIs are summarized in Table XII. The values of different criteria of the original SpiralMamba are exhibited for comparison. Also, the two models' OA, AA, and Kappa values are displayed for reference. Unsurprisingly, spiral scan increases the computational complexity in all cases. The reason is that there are eight scanning paths in spiral scan, while flatten only has one. However, we believe this growth is worthwhile due to the two following points. First, the increase in complexity is not dramatic. Compared to SpiralMamba with flatten, the metric increases caused by spiral scan are acceptable. For example, the Params of SpiralMamba with flatten and spiral scan counted on HUS2018 are 0.14 and 0.15 MB. Considering the eightfold increase in the scan path, the growth in these parameters is minimal and tolerable. Second, the multipath spiral scan provides higher performance gains. The OA, AA, and Kappa values of SpiralMamba with spiral scan are higher than that of SpiralMamba with flatten. For instance, compared with flatten, the enhancements in OA achieved by spiral scan are 0.20% (IP), 0.24% (PU), 0.51% (HUS2013), and 1.30% (HUS2018).

3) MLC Ablation: To further prove the effectiveness of the MLC, we conduct the following experiments. First, three models are constructed, and they are named "with None," "with MLP," and "with MLC." Here, "with None" means that we remove MLC from SpiralMamba, "with MLP" indicates that MLC is replaced by MLP, and "with MLC" is our SpiralMamba. Their performance counted on different HSIs is summarized in Table XIII. By observing the results, we can easily find that the behavior of "with None" is the weakest. This indicates that the lack of spectral token internal information extraction in SpeME reduces the model's classification performance. When adding MLP and MLC, the models' behavior is strengthened, and "with MLC" outperforms "with MLP" in all cases. This highlights that MLC is more suitable for modeling within spectral tokens than MLP. MLP struggles to capture the local correlations within spectral tokens as a global component. In contrast, MLC using convolution can more finely perceive local variations.

4) Fusion Module Ablation: To further verify the effectiveness of SSCFM, we perform additional experiments, including the single branch and the dual branch using addition or concatenation fusion. The experimental results are listed in Table XIV. It can be observed that the SpeME only performs better than the SpaME only on the IP dataset, indicating that spectral features are learned more effectively than spatial features. For the dual-branch fusion, the performance of both addition and concatenation fusion is similar, and both outperform SpaME only but are inferior to SpeME only. This suggests that simple addition or concatenation fusion does not significantly improve performance. On the PU, HUS2013, and

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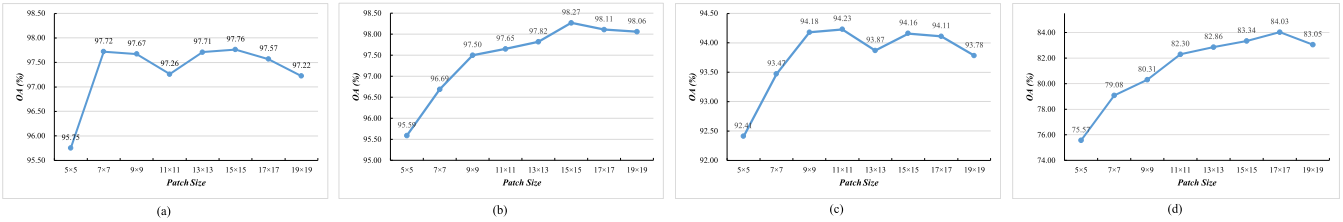


Fig. 12. OA curves of different SpiralMambas with various patch sizes on four datasets. (a) IP. (b) PU. (c) HUS2013. (d) HUS2018.

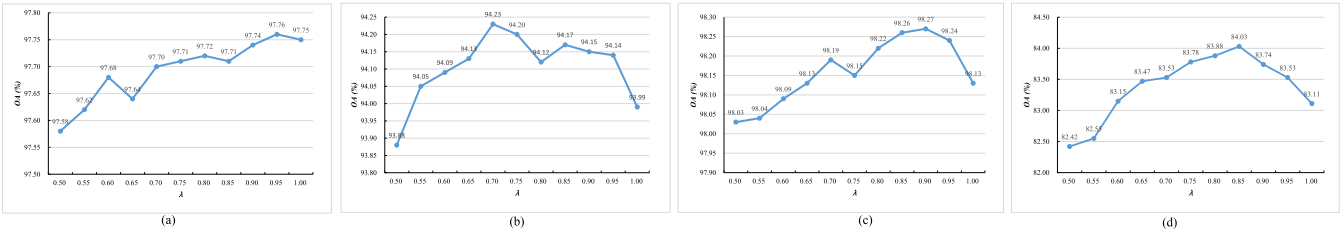


Fig. 13. OA curves of different SpiralMambas with different λ on four datasets. (a) IP. (b) PU. (c) HUS2013. (d) HUS2018.

HUS2018, SpaME outperforms SpeME only, implying that spatial features are more discriminative for these datasets. In the case of dual-branch fusion, while the addition and concatenation fusion show improvements over single branch, there is still room for further enhancement. The dual-branch fusion with SSCFM consistently performs best across all four datasets. This is because SSCFM effectively captures the complex relationships between spatial and spectral features by RLB and uses TEM to enhance high-confidence features while filtering out low-confidence features, resulting in more accurate fusion.

E. Sensitivity Study

As mentioned in Section IV-B1, two free parameters should be set in advance, they are the patch size and threshold λ. In this section, we study their influence in detail. For clarity, we only count OA values here.

1) Impact of Patch Size: To investigate the impact of patch size on SpiralMamba, we vary it from 5×5 to 19×19 and count the results of different SpiralMambas on four datasets. The results are shown in Fig. 12. Overall, as patch size increases, the OA first increases and then decreases for all four datasets. This trend can be attributed to smaller patch sizes containing less spatial context information. As the patch size increases, spatial information becomes richer. However, when the patch size is too large, more noise and redundant information may be introduced to hinder the model's learning. Additionally, it can be observed that the best performance on the IP, PU, HUS2013, and HUS2018 datasets is achieved with patch sizes of 15×15, 15×15, 11×11, and 17×17, respectively. This suggests that SpiralMamba performs better with larger patch sizes, demonstrating its strong capability for long-range dependency modeling. Even with a larger patch size, it can efficiently capture spatial–spectral clues and achieve satisfactory classification results.

However, there is a non-negligible fluctuation phenomenon (patch size 11×11 for IP and 13×13 for HUS2013). In IP, the distribution of land cover categories is relatively regular and concentrated, and a larger field of view enables a better overall understanding of the targets. Meanwhile, our SpiralMamba, which has strong global perception capabilities, tends to favor larger patch sizes. When the patch size is 11×11 (the fluctuation point), SpiralMamba's field of view is not large enough, potentially failing to fully activate its global modeling ability, leading to suboptimal performance. As the patch size increases, the field of view becomes broader, improving performance, with the optimal result appearing at 15×15. In HUS2013, the land cover is more complex, with a more redundant background. As patch size increases, the model's performance peaks at 11×11. However, as it increases further to 13×13 (the fluctuation point), excessive redundant information may be introduced, leading to a decline in performance. Although performance slightly improves at 15×15 (likely because useful spatial information outweighs redundant information at this size), it still remains below the peak at 11×11 due to the continued impact of redundancy, and the performance shows a downward trend thereafter.

2) Impact of Threshold λ: To study the effect of the threshold λ on our model, we change its values from 0.5 to 1 with an interval of 0.05. The performance of SpiralMamba with different λ counted on four datasets is shown in Fig. 13. The OA curves show a trend of increasing first and then decreasing as λ increases, with peak values occurring at 0.95 (IP), 0.90 (PU), 0.70 (HUS2013), and 0.85 (HUS2018). When λ is set to its minimum value of 0.5, all values in the relationship weights are filtered (set to 0) or enhanced (set to 1) by the TEM, resulting in a simple combination of spatial and spectral features and the lowest OA values. When λ exceeds 0.5, the obtained features contain spatial, spectral, and spatial–spectral messages, and the proportion of spatial–spectral clues would be increased as λ grows. This integration pushes the models to achieve optimal performance. When λ reaches 1, TEM will not refine the relationship weights. Unreliable features may contaminate reliable features during fusion, and accurate

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TABLE XV
PARAMETER SIZE, MEMORY USAGE, AVERAGE TRAINING TIME PER EPOCH, AND INFERENCE TIME OF ALL METHODS ON IP

Methods	Params (*MB*)	Memory (*MB*)	Training (*s/epoch*)	Inference (*s*)
SpectralFormer [65]	0.35	1190	0.50	4.09
MorphFormer [66]	0.27	1392	0.65	3.72
SQSFormer [68]	1.32	4033	0.97	5.56
GraphGST [67]	2.07	**672**	0.73	4.54
DBCTNet [69]	**0.03**	1476	2.67	5.90
HSIMamba [57]	26.0	894	3.68	1.81
S^2Mamba [54]	0.87	1408	0.41	1.82
MiM [53]	1.70	5622	2.34	5.23
MambaHSI [58]	0.42	2562	**0.12**	**0.05**
MambaLG [59]	0.19	2376	0.40	0.61
IGroupSS-Mamba [60]	0.06	704	1.28	4.00
SpiralMamba(Ours)	1.97	1742	0.70	2.10

fusion cannot be achieved. Therefore, the behavior of the model becomes weaker.

F. Complexity Analysis

To better understand SpiralMamba's complexity, we count its Params, Memory, Training, and Inference on the IP dataset and summarize them in Table XV. The results of the compared methods are also exhibited for reference. To be fair, the HSI patch size is fixed as 15×15 for all models, consistent with SpiralMamba. From Table XV, we can observe the following points. First, compared to the methods with similar parameter sizes (i.e., SQSFormer, GraphGST, and MiM), SpiralMamba not only has the shortest average training time per epoch but also cuts inference time by half. Additionally, it demonstrates better memory efficiency (except for GraphGST, which uses fewer nodes in its graph structure, resulting in the least memory usage). This shows that our model is more efficient with a similar parameter size. Second, compared to all Transformer-based methods, SpiralMamba, despite having more parameters, achieves comparable memory usage and training time while consuming the least inference time. This is due to the efficient Mamba structure. It has linear complexity with the same sequence length, which leads to significant time savings. Furthermore, all Mamba-based methods have faster inference times than Transformer-based ones, further highlighting Mamba's efficiency.

V. CONCLUSION

This article proposes a new HSI classification method based on Mamba, called SpiralMamba. It has a dual-branch fusion architecture and aims to fully explore spatial context and rich spectral features from HSIs for classification. First, a SpaME is developed to mine spatial knowledge. SpaME introduces a spatial spiral scan strategy to traverse HSI pixels effectively at the spatial aspect. This way, HSI can be transformed into a reasonable sequence for SSM, thus enhancing SpiralMamba's ability to understand spatial context. Second, a SpeME is constructed to capture spectral clues. Besides SpeBiM, which focuses on analyzing the spectral relations, an MLC is introduced to capture local variations within spectral tokens, compensating for the limitations of SpeBiM in modeling within the spectral token itself. In this fashion, SpiralMamba is able to comprehensively model complex spectral features, leading to discriminative spectral representations. Finally, SSCFM is designed to complete the spatial–spectral information fusion. Unlike common modules, it first deeply mines the intricate relationships between the two features. Then, the generated relationship weights are refined by TEM to ensure the spatial and spectral features can be considered thoroughly. Extensive experiments counted on three publicly available datasets validate the effectiveness of SpiralMamba.

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